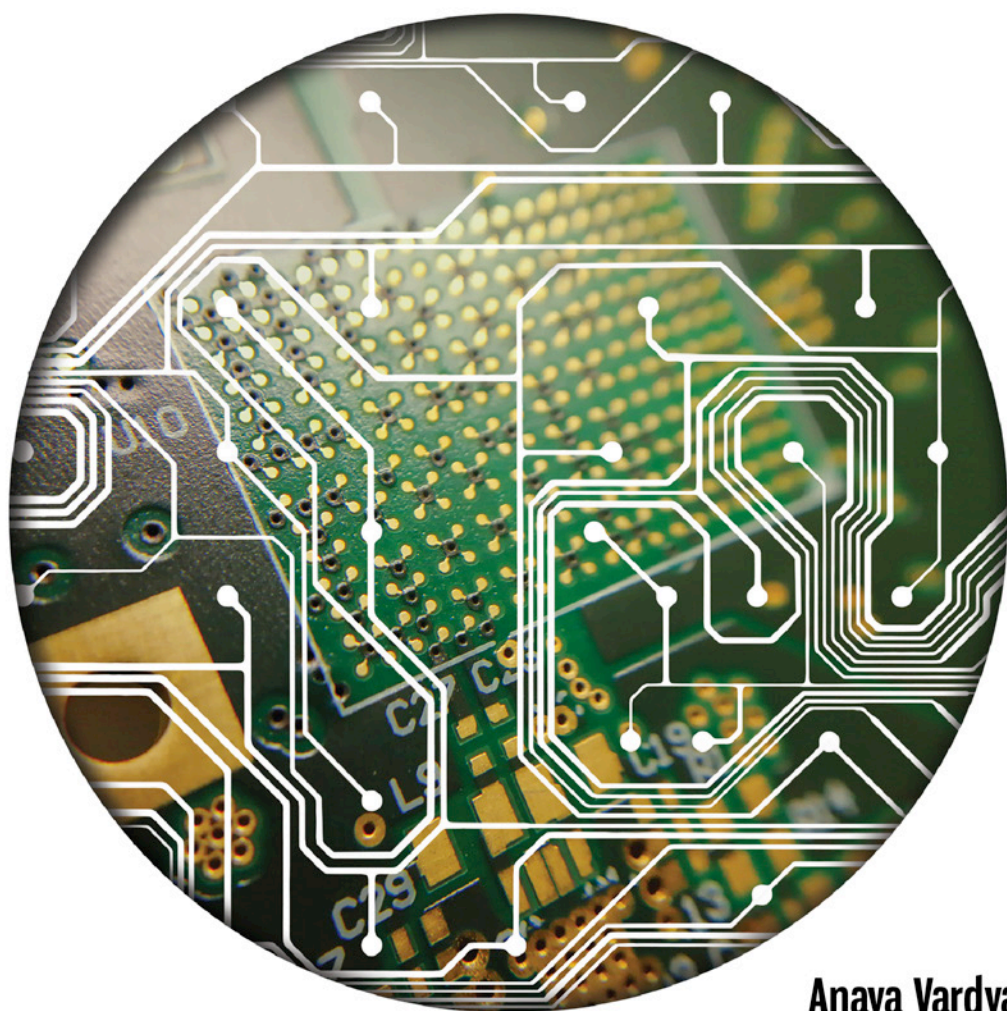


THE PRINTED CIRCUIT DESIGNER'S GUIDE TO...™

DFM Essentials



Anaya Vardya
American Standard Circuits
ASC Sunstone Circuits

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Books

The Printed Circuit Designer's Guide to...™ DFM Essentials

Anaya Vardya

American Standard Circuits

ASC Sunstone Circuits

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PEER REVIEWERS

This book has been reviewed for technical accuracy by the following experts.



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Currently the principal consultant at Korf Consultancy LLC, Dana Korf works with companies to improve PCB fabricator front-end engineering processes. Dana is AME standards manager at Nano Dimension. He also works with OEMs and ODMs to create design rules and technology roadmaps, assist with supplier selection/qualification, and reduce DFM cycles. Dana previously worked in China for over seven years at Multek in Zhuhai as senior director of manufacturing engineering and NPI, and at Huawei Technologies in Shenzhen as director of PCB technology where he was responsible for PCB technology ranging from mobile phones to RF antennas, base stations, and high-speed digital servers and switches. Before that, he worked for Samina-SCI, HADCO, and Zycon as director of product engineering. Dana has been awarded the IPC President's Award, chaired many high-speed IPC committees, and was a co-chair for the iNEMI Data Convergence project, which has become IPC-2581. Dana graduated from Washington State University with a BSEE, and he enjoys following college and professional football and golf.



Kris Moyer

Instructor, IPC

Kristopher Moyer is a distinguished PCB design expert with over 30 years of experience, boasting a portfolio of over 1,000 PCB designs. With an educational background that includes an AS in electrical engineering from ITT Technical Institute and a BS in computer engineering from California State University, Sacramento (CSUS), Kris has solidified his expertise in the field. Notably, he continues to serve as part-time faculty at CSUS, where he shares his knowledge with the next generation of engineers and is an IPC-certified master printed circuit board designer.

Kris's remarkable career is highlighted by his leadership roles within the IPC design community, including co-chairing key committees focused on land pattern and high-frequency design guides. His work extends to significant projects in military and space applications, demonstrating his capability to lead complex design endeavors, including a notable 24-layer board project.

Kris's technical proficiency encompasses a deep understanding of FPGA development, mastery of design and simulation tools, and a strong foundation in environmental testing for PCBs. Beyond that, he is recognized for his contributions to PCB design standards and practices through his active involvement in numerous IPC committees and his role as an instructor and subject matter expert with IPC.

MEET THE AUTHOR



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Anaya Vardya has over 40 years' experience in the electronics manufacturing business and is currently the president and CEO of American Standard Circuits LLC and CEO of Sunstone Circuits LLC. Anaya was previously COO of Canadian-based Coretec Inc. and senior VP of operations at Merix (both now part of TTM). He was also the corporate development manager at Continental Circuits, and before that, he held a variety of management and non-management positions in IBM Endicott and IBM Austin. Anaya holds a master's degree in chemical engineering from the University of Cincinnati and a Bachelor of Technology from the Indian Institute of Technology.



American Standard Circuits

Sunstone Circuits

Contents

1 **Introduction**

Chapter 1:

3 **Materials**

Chapter 2:

9 **Panelization**

Chapter 3:

13 **Controlled Impedance**

Chapter 4:

19 **Material Construction Stackups**

Chapter 5:

25 **Via Structures**

Chapter 6:

31 **Plating Methods**

Chapter 7:

37 **Solder Mask and Legend**

Chapter 8:

41 **Final Finishes and Design Guidelines**

Chapter 9:

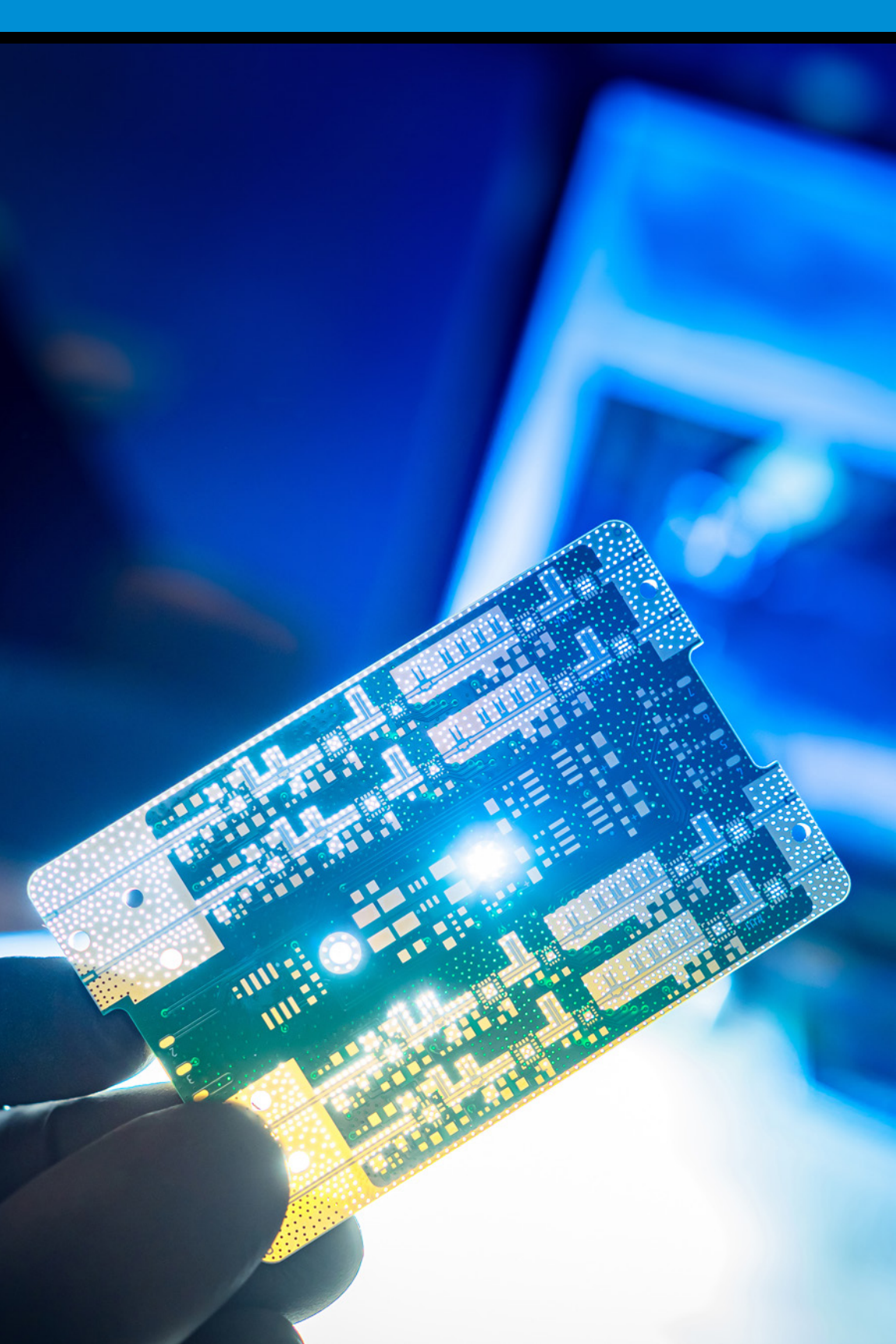
53 **Data Package Guidelines**

Chapter 10:

57 **Cost Driver Summary**

60 **Conclusion**

63 **About American Standard Circuits**



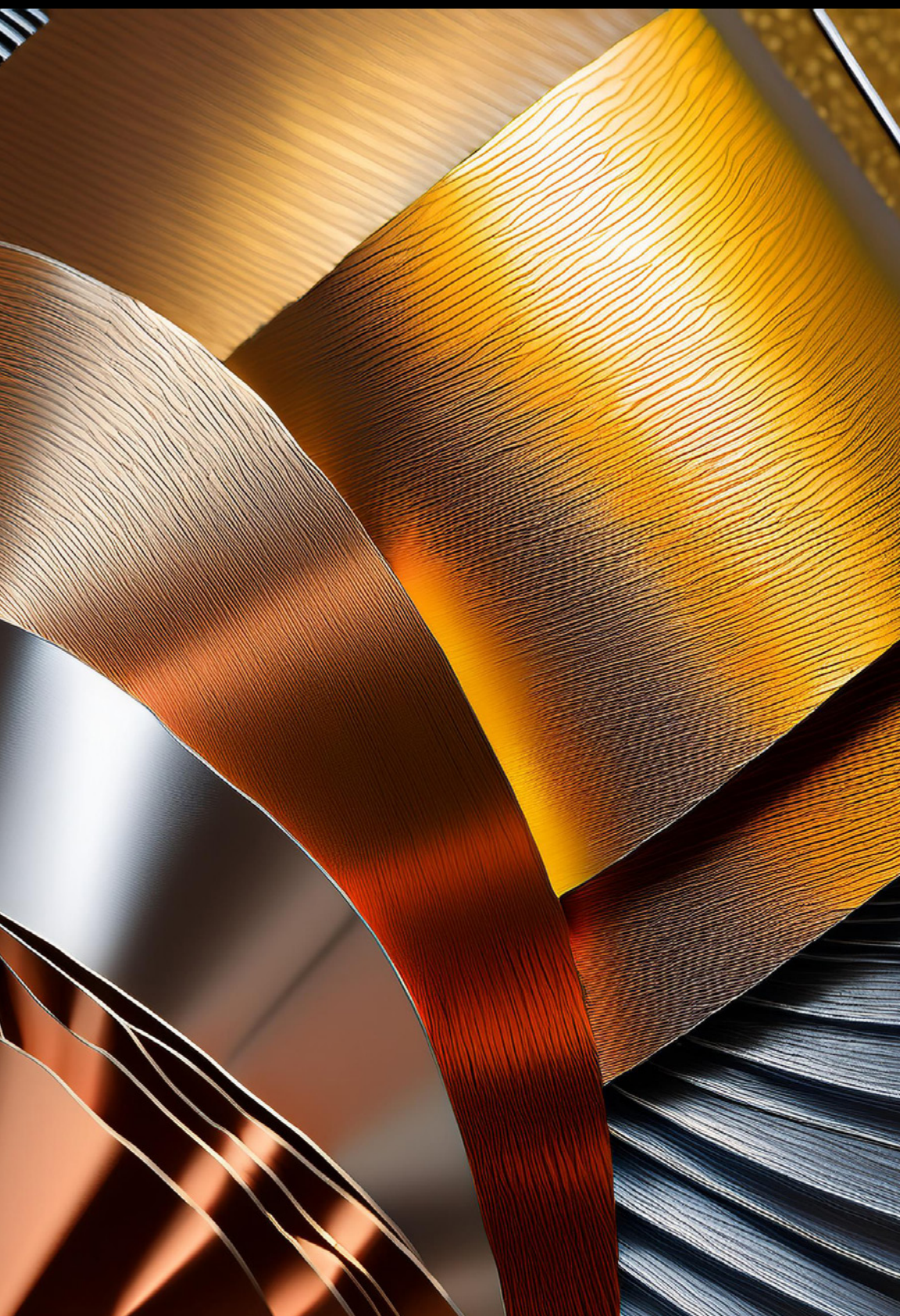
Introduction

One of the biggest challenges facing printed circuit board designers is properly understanding the cost drivers in the PCB manufacturing process, particularly the manufacturing of advanced technology PCBs.

Design for Manufacturing (DFM) is defined as the practice of designing printed circuit boards that meet not only the capabilities of the customer's assembly manufacturing process, but also the capabilities of the board fabrication process to provide the highest reliability at the lowest possible cost.

The guidelines offered in this book are based on both ASC recommendations and IPC standards and apply to standard to advanced technology PCBs. UHDI technology may require violating many of the guidelines included within. It is also understood that some of these guidelines may require adjustment based on the material set, fabricator processes, and other design constraints.

While not a substitute for early design engagement with the PCB fabricator, in this book the reader will learn about these cost drivers (from the PCB manufacturer's perspective) and the design decisions that will impact product reliability and help companies "design for success."



Materials

Standard Multilayer Materials

Most PCBs are manufactured using three basic materials: glass-reinforced epoxy laminate, prepreg, and copper foil. PCBs are constructed from three basic material types: copper foil, prepreg, and cores.

Copper foil: Sheets of copper foil are incorporated into the outer layer of the PCB, laminating it onto the prepreg to create the outer layers. Outer layers are generally constructed using ½-ounce copper or thicker depending on the design requirements. Internal layers are constructed with copper that is specified on the fabrication print. Half-ounce copper foil is commonly used for signal layers; 1 ounce for plane layers, and 2 ounce or greater for power planes where there is a high DC current. Other thicknesses may be used based on the design requirements.

Prepreg: This is a semi-cured glass resin material. The resin used for the FR-4 type materials is epoxy-based. There is no copper attached to this material.

Core: This is fully cured glass-resin material with copper laminated to both sides. This is typically used for internal layers. It is occasionally used for outer layers, but that is not a preferred construction method. A core is constructed from either one sheet of prepreg (single ply) or two or more sheets of prepreg, and two layers of copper foil. Single ply is considered the preferred core construction and has better dimensional stability.

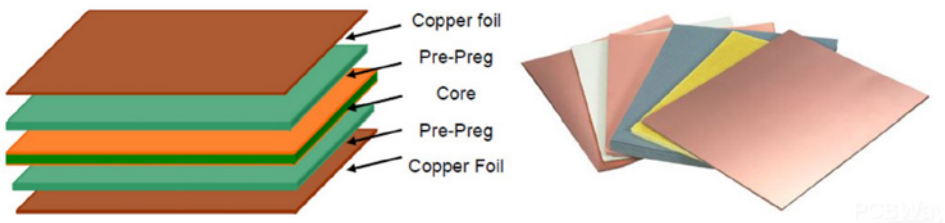


Figure 1.1: Typical 4-layer material construction.

High-frequency (RF/Microwave) Materials

High-frequency designs (1 GHz and up) require materials with closely controlled dielectric constants and dissipation factors. The FR-4 materials normally used for PCBs don't have the desired controlled characteristics. The substrate mate-

rials used for high-frequency applications were originally based upon PTFE resin formulations that have the desired properties, i.e. dielectric constant (DK) controlled to ± 0.04 and dissipation factor (Df) to 0.0004. These values may vary somewhat depending on the material type and supplier. Today, there are a number of materials on the market that do not contain PTFE resin but still have controlled values that can be used for high-frequency applications.

Material Types

Non-woven materials contain a dispersion of glass microfibers in the substrate. These are typically materials with low dielectric constants (2.20–2.35). They work very well at the higher frequencies, as the dissipation factor is low.

Woven glass materials are made using fine weave glass cloth. These materials have dielectric constants in the range of 2.40 to 2.60. Due to the glass cloth, the dissipation factor may be somewhat higher.

Filled materials have dielectric constants ranging from 2.94 to 10.8. The filler material may be a ceramic or other suitable material that can be used to raise the dielectric constant. These materials may also contain non-woven or woven glass. The type of filler and construction determines the dielectric constant as well as the dissipation factor. These materials tend to have a higher dissipation factor than the materials mentioned above.

Copper Foil Types

Two types of bonded copper foil are available: electro-deposited (ED) and rolled-annealed (RA). The difference between these foils is the fabrication

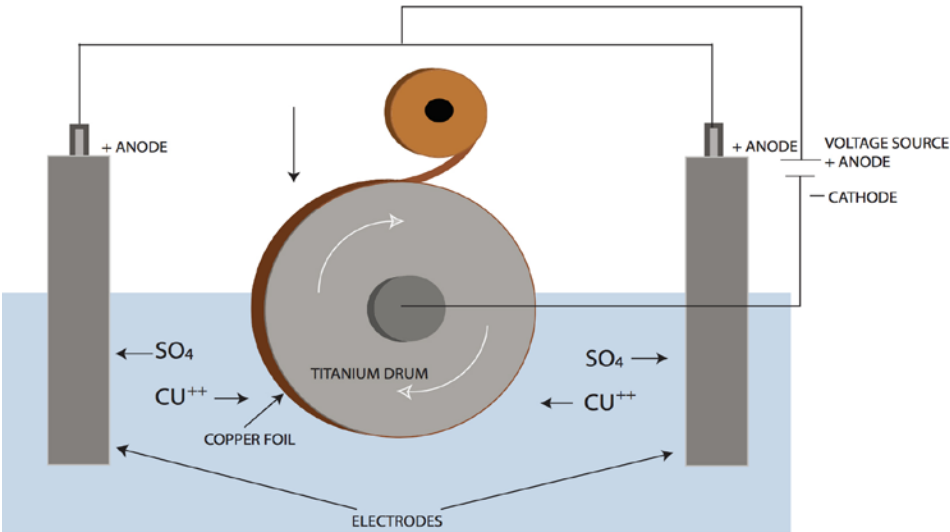


Figure 1.2: Electrodeposited (ED) copper foil drum process.

process and the treatment on the backside to enhance bonding adhesion.

Electrodeposited copper (ED) is made by electroplating onto a rotating drum that results in a vertically oriented grain structure. Opposite the drum surface, the copper plating is rough—almost like micro teeth—which is enhanced by further treatment. The “teeth” provide higher peel strength but may increase conductor losses at high frequency because of the depth of the teeth.

Rolled-annealed copper (RA) is made by rolling an ingot into very thin foils. The grain structure is horizontally oriented. The rolling causes stress in the foil and must be heat annealed to remove the stress. Rolled-annealed copper has the same rough finish (teeth) as described above in the ED drum process, but the teeth are on both sides of the foil. However, rolled-annealed copper has about half the peel strength of electrodeposited copper. This means rolled-annealed copper performs better at frequencies above 13 GHz and is a primary candidate for flex technology. Grain direction needs to be considered when engineering stackups because the rolling process induces stretching in the copper foil in one direction.

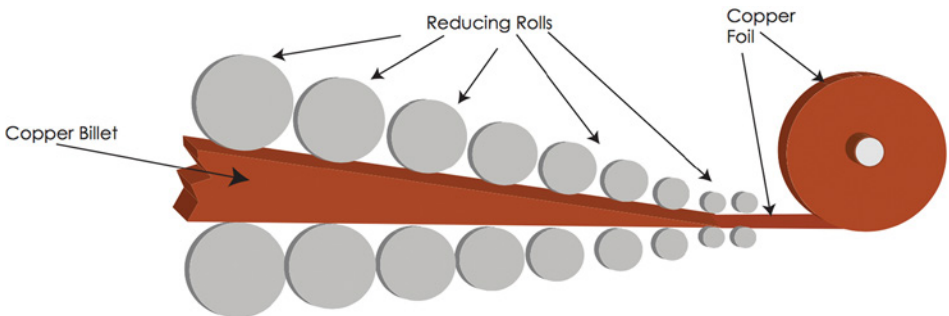


Figure 1.3: Rolled-annealed (RA) copper foil process.

Peel strength is the amount of force, in pounds per inch, which must be applied to cause a 1-inch (2.54 cm) wide strip of the copper foil to peel from the substrate. Typical values for electrodeposited 1-ounce ED copper are 9–11 pounds (4.1–5.0 kg) per inch; for 1-ounce RA copper typical values are 4–6 pounds (1.8–2.7 kg) per inch. As circuit traces become narrower, the peel strength is reduced. Therefore, more care is required to solder and bond to narrow traces.

Copper surface roughness is critical for high-frequency applications where the roughness of the copper surface impacts conductor loss. At high signal frequencies, the current in PCB copper circuit is concentrated within a small depth near its surface, referred to as the “skin effect” or “skin depth.” Skin depth is a measure of how (and where) electrical conduction takes place in a conductor and is a function of frequency.

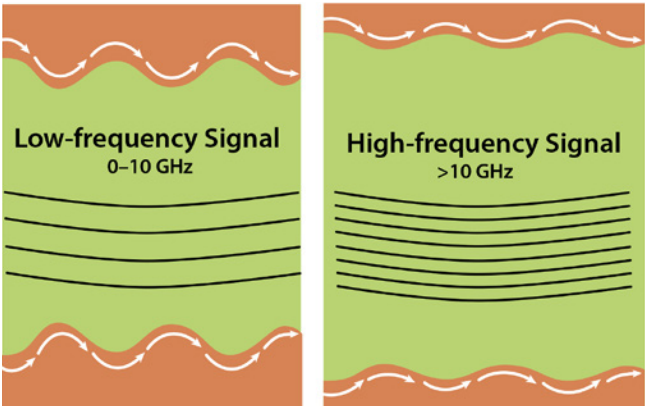


Figure 1.4: As signal frequency increases, more electrical signal is lost due to increased resistance from more closely following the irregularities in surface copper "skin" roughness.

In other words, for high-frequency applications, the smoother the surface copper, the better. Circuit conduction occurs at the surface nearest the dielectric from which the EM wave propagates, i.e., the bottom copper surface against the laminate in a microstrip design.

Metal Core and IMPCB Materials

Metal core or insulated metal printed circuit boards (IMPCBs) are thermal management designs that utilize a layer of solid metal to dissipate the heat generated by the various components on the board. When metal is attached to a PCB, the bonding material can either be thermally conductive, but electrically isolative or, in the case of RF/microwave circuits, it may be both electrically and thermally conductive. There are two common metals used in these applications: copper and aluminum. Which to use depends on the application with consideration for cost, weight, and thermal properties. It should also be noted that a secondary use of metal core boards is to add stability and stiffness in designs where stiffeners cannot be added to the outer layers. Table 1.1 presents the properties of the various base metals.

Table 1.1: Properties of various base metals.

Metal Base Material	Thermal Conductivity (W/mK)	Thermal Expansion (ppm/K)	Comments
Aluminum 5052 H32	138	25	Al-Mg-Cr alloy: Best for bending and mechanical forming, most popular choice, low cost.
Aluminum 6061 T6	167	25	Al-Mg-Si-Cu alloy: Best for CNC machining and V-cut scoring, medium cost.
Copper C110	386	17	Pure Cu: Low CTE, high thermal conductivity, high cost.

Embedded Capacitance Materials

Embedded capacitance material consists of a very thin layer (≤ 2 mils) of ceramic-filled FR-4 or polyimide epoxy or other specialty dielectric material sandwiched between two layers of copper foil. Some materials can also include a glass filler. It can be patterned as a power and ground plane pair for a lower impedance, shared capacitance power distribution network, or as discrete capacitors to replace physical capacitors from the surface of the board assembly. The biggest design benefit is that it increases the usable board area by allowing for the removal of many, if not all, decoupling capacitors equal to or below $0.1 \mu\text{F}$ and their associated solder joints and vias. Additional benefits include:

- Better power delivery
- Lower profile
- Low inductance and impedance
- Thickness and weight reduction
- Higher reliability

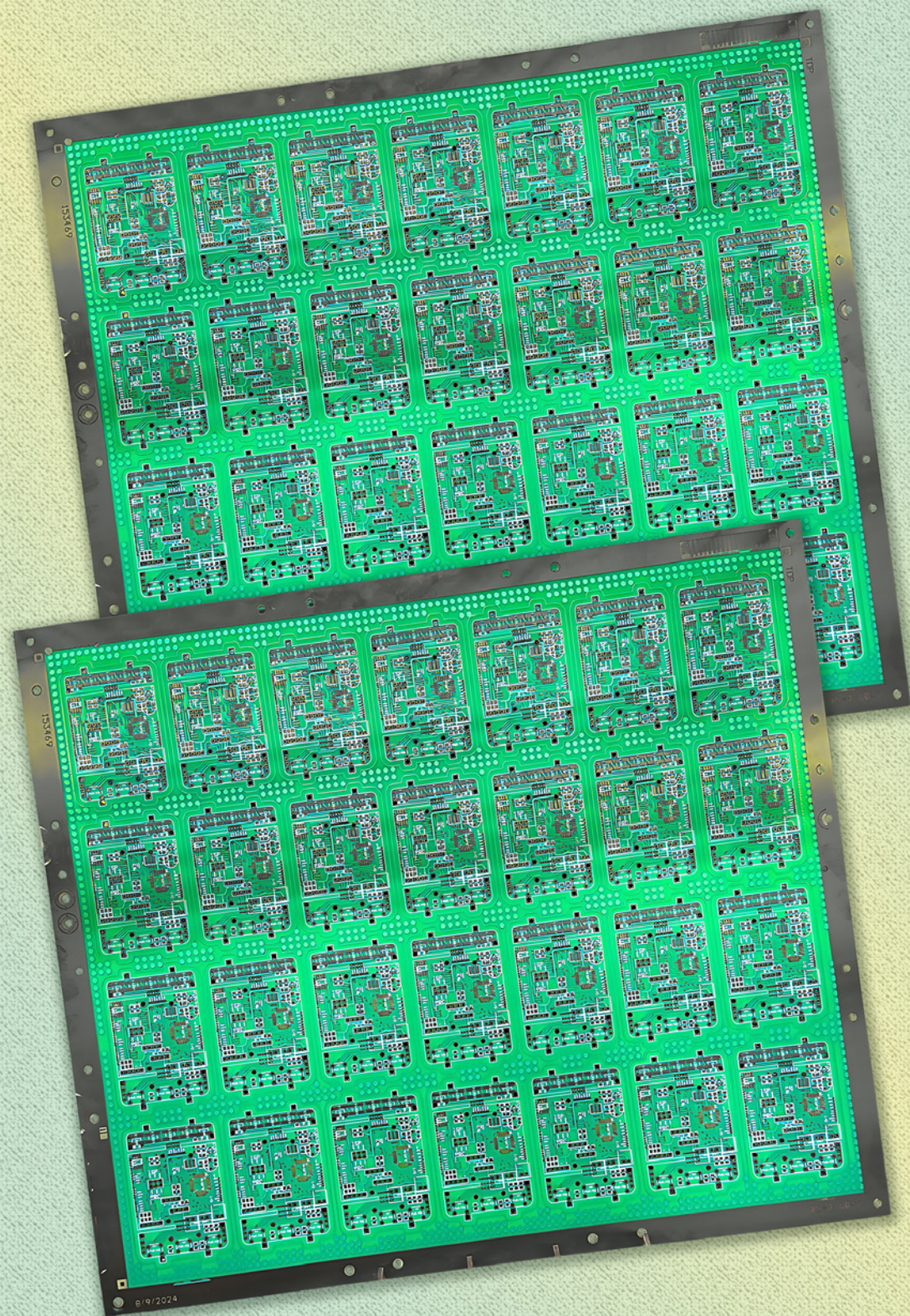
Embedded Resistor Materials

Embedded resistors are planar resistive elements made into a thin film. This type of resistor becomes part of the etched and printed circuitry on the standard printed circuit board layer as it eliminates the need for solder joints.

Additional benefits:

- Improves line impedance matching
- Reduces signal paths and series inductance
- Reduces crosstalk, noise, and EMI
- Enhances performance of cost-effective resistor technology in high-speed, high-density designs
- Eliminates discrete resistors
- Reduces assembly time
- Stable through a wide frequency range of 20+ Ghz

Multilayer PCBs use a variety of materials to achieve a specific functionality for a specific application with new material sets and formulations continually being introduced. The choice of materials depends on factors such as the application, frequency of operation, cost considerations, and manufacturing requirements. It is always best to engage early and often with your PCB fabricator to ensure the correct material set for any given design and application.



Panelization

Panelization is the process of placing one or more PCBs on a manufacturing panel and incorporating features to assist manufacturing (such as tooling holes, fiducials, coupons, panel thieving, etc.). Panelization is one of the highest impact factors in the cost of a PCB.

The panel area available for circuit boards plus coupons is known as the usable area (Figure 2.1). Test coupons are added by the fabricator to the perimeter of the manufacturing panel outside the usable area. The number and type of coupons are based on the specs that the PCBs are being built to and requirements for controlled impedance. Coupons may sometimes further reduce the amount of a PCB panel that is available for the circuit boards. This is defined by the fabricator as required perimeter for coupons. This is typically about 1 inch or 25.4 mm and is a fixed area based on panel size. Panel utilization is measured as a percentage, defined by the total area for PCBs divided by the total panel area.

Cost-effective material utilization is defined by a target panel with greater than 75% panel utilization.

Raw laminate is one of the primary cost constituents of a multilayer PCB. Optimizing panel structures around standard base materials while achieving maximum material utilization on one of the standard panel sizes can have a significant impact on multilayer board prices and deliveries. The three most preferred sizes in North America are 12x18 inches, 18x24 inches and 21x24 inches, with 18x24

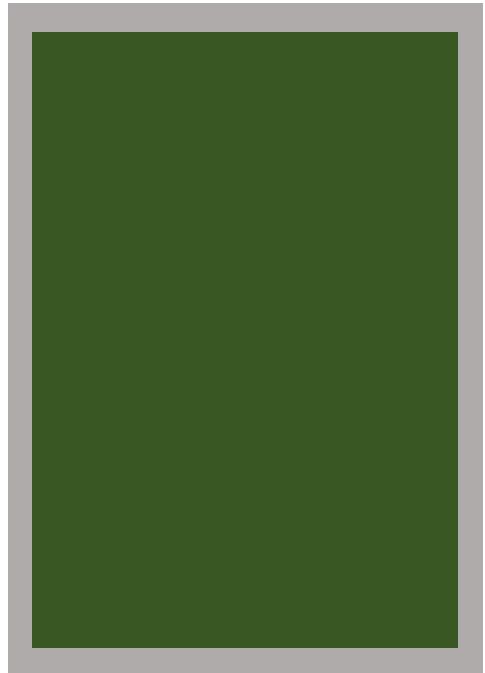
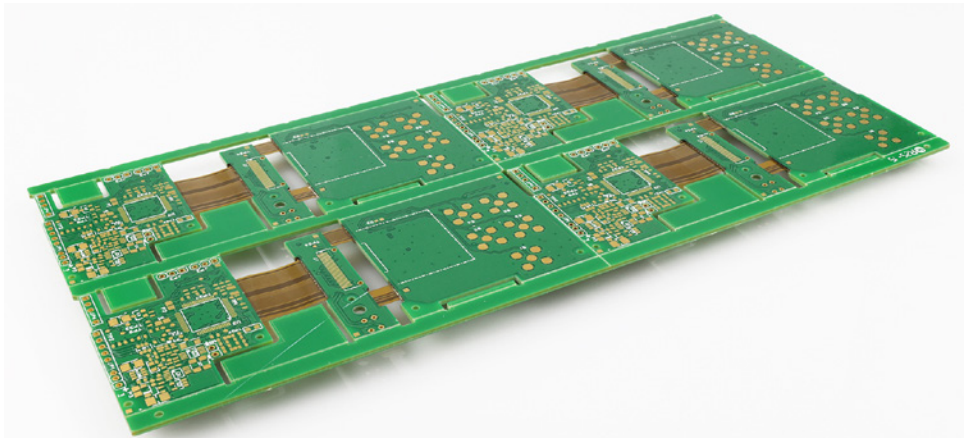


Figure 2.1: Typical panel with border shown in gray; usable area in green.



inches being the most common. For special applications, other panel sizes can be provided. In Asia, typically the shops will process custom panel sizes in order to maximize utilization. The most effective cost per unit area processed is usually found with a larger panel size.

The most effective material utilization will be achieved with PCBs or arrays of PCBs such that their finished outline fits as efficiently as possible within the useable area of the panel. Test coupons may be within the useable area, but are typically placed in the manufacturer perimeter mentioned previously. Material utilization may be increased by employing the scoring process. This process places grooves on opposite sides of the panel between boards for snapping the boards from the panel. This method allows boards to be butted up against each other, possibly eliminating the real estate for rout paths and allowing more boards to be placed on the panel.

For single- and double-sided product, a 0.500" (1.270 cm) border is required around the periphery of the panel for tooling purposes. For multilayers, this allowance is 0.750" (1.905 cm). Higher technology will require larger borders, typically 1.5" (3.81 cm) or greater to accommodate coupon and other testing/verification features.

Panel Useable Area

The following four general modifications to a panel will reduce the available useable area. Borders can vary by manufacturer, it is best to inquire about specific constraints.

Step-and-repeat

This is the process of reproducing successive images onto a panel. For PCBs without gold-plated edge contacts, the standard step-and-repeat spacing between routed parts is normally 0.100" (2.54 mm) and 0.0 spacing when scored.

Controlled Impedance Coupons

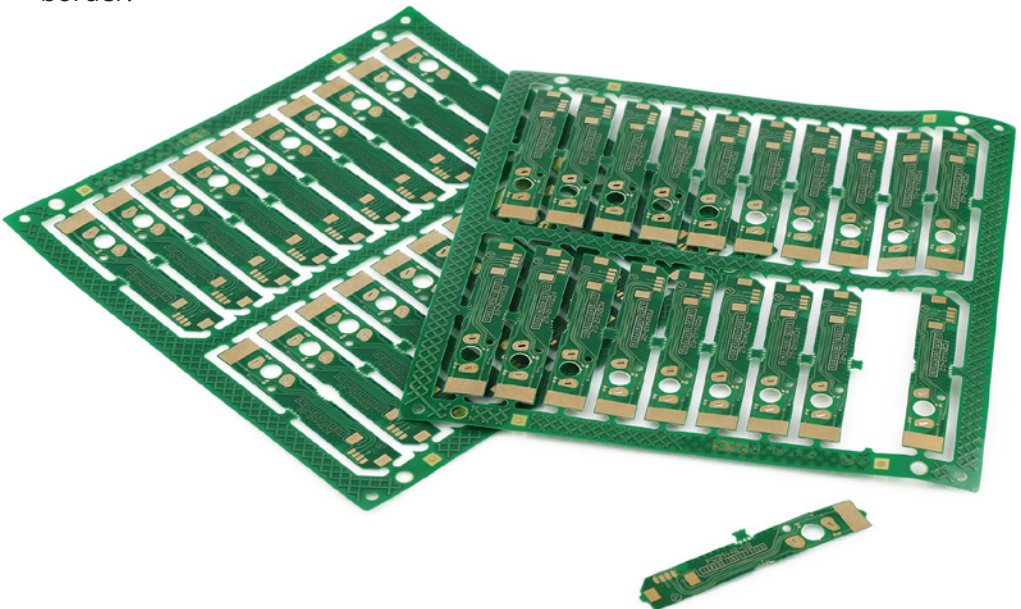
PCBs with controlled impedance technology are processed with test coupons as part of the lay-up. Typically, an additional 1" (2.54 cm) border area is required. When 20% or less of nominal impedance tolerances is specified, using controlled geometry coupons is recommended to ensure values are met.

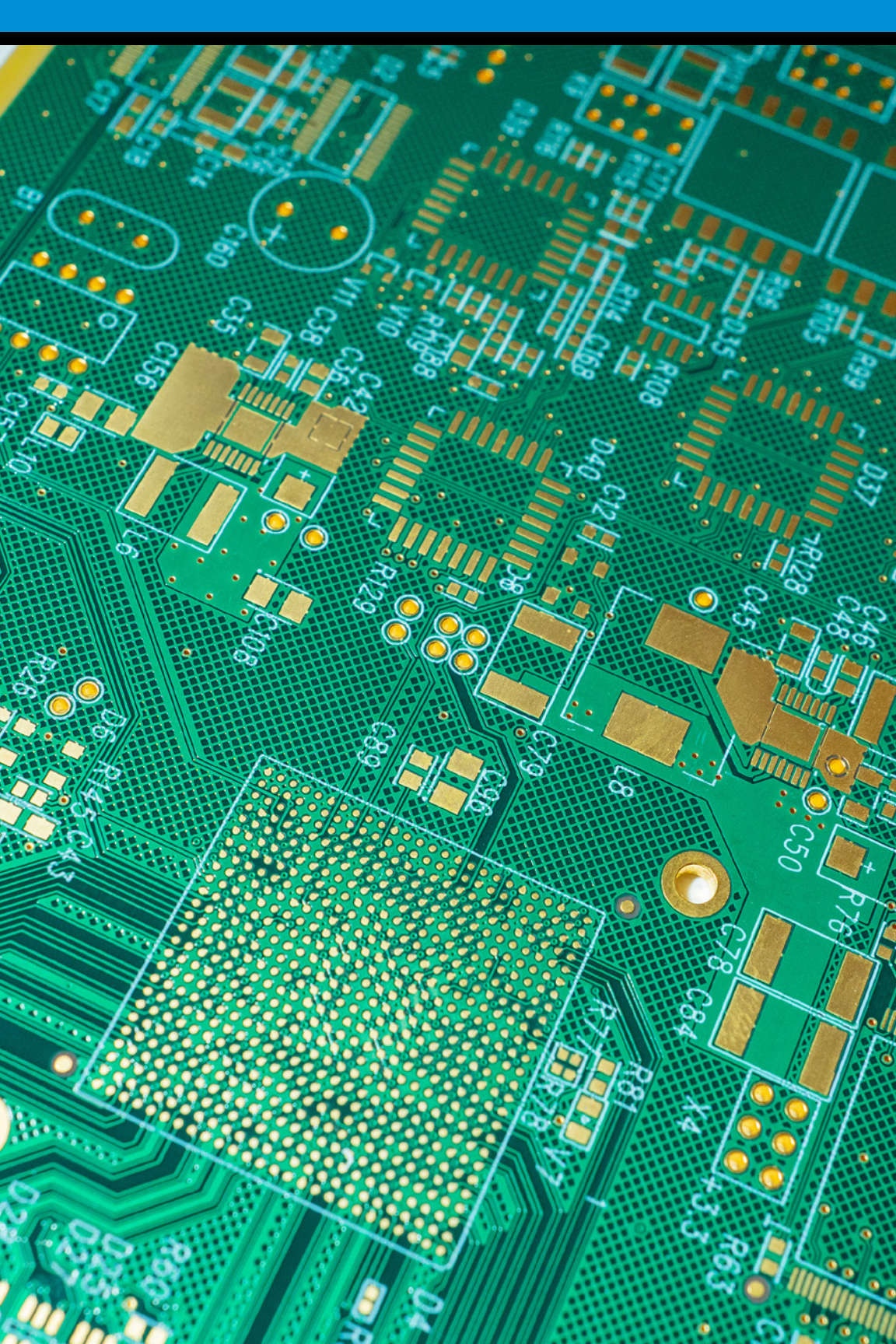
MIL-SPEC Coupons

Any PCBs requiring MIL-spec conformance will require additional MIL-SPEC coupon(s) be placed on each PCB. Depending on the individual fabricator's coupon designs, additional border area may be required.

Assembly Rails

A minimum border of 0.375"-0.5" (9.525 mm-12.7 mm) typically must be maintained around the perimeter of the panel to facilitate assembly and may vary by contract manufacturer. It is a common practice to allow the assembly rail (breakaway) to protrude into the border to keep the PCB away from the border.





Controlled Impedance

Printed circuit boards that contain controlled impedance traces require specific constructions and tighter manufacturing process controls. The fabricator needs to tailor the construction for PCBs requiring impedance to precisely match the required nominal impedance values. The fabrication drawing should specify the required nominal impedance and tolerance and the fabricator will create a construction to meet the intended design impedance requirements. Routing preference on internal layers for controlled impedance traces must consider the difference in structure and Dk value for the environment (air) above the trace on an outer layer and absence of reference plane above the trace for a microstrip/embedded microstrip compared to the uniform structure of stripline/dual stripline.

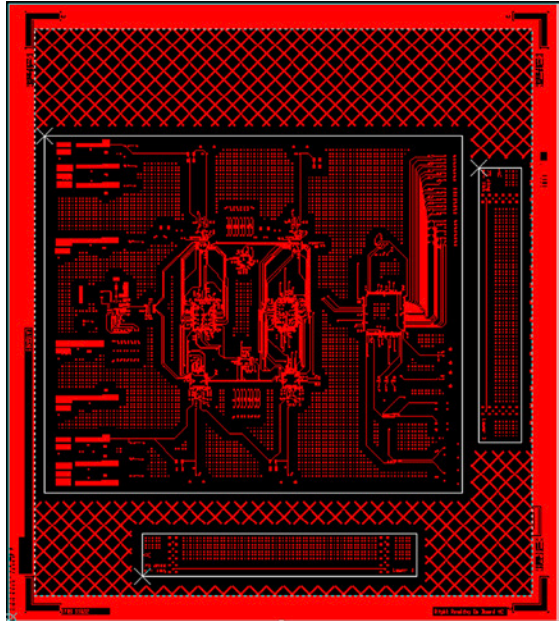


Figure 3.1: CAM screenshot of a typical controlled impedance design.

Primary Impedance Factors

- Trace width
- Copper thickness
- Dielectric spacing
- Dielectric constant (Dk)/relative permittivity (ϵ_r)
- Indicated "Reference Only" (See drawing/specification impedance requirements)
- Material requirements

Fabricators use some form of impedance modeling software to determine the specific PCB construction required to produce the specified impedance. The PCB drawing should specify the nominal impedance, tolerance, and nominal line width. This will allow for the creation of the most cost-effective PCB material construction.

Multiple Impedance Considerations

Some PCBs require multiple impedance values on the same signal layers. Your PCB fabricator must have the ability to create impedance coupons to reflect the appropriate model for each impedance requirement. However, testing multiple impedance values on a given signal layer could require coupon redesign and may cause the coupon to be wider than normal and reduce the panel utilization.

There are two types of impedance structure models that are generally specified: Single-ended and Differential.

Single-ended Impedance

Single-ended impedance is established by the interaction of a single trace and its reference plane(s). There are four basic single-ended impedance structures:

1. **Microstrip:** a trace on an outer layer with a single reference plane below it.
2. **Embedded microstrip:** a microstrip line that has a dielectric over the top of it; solder mask will change a microstrip into an embedded microstrip line, although the impact is negligible.
3. **Stripline:** a trace on an internal layer that has a reference plane above and below it.
4. **Dual stripline or offset stripline:** a stripline, which is offset between the two reference planes; it generally is used when two adjacent signal layers are routed orthogonally and have reference planes outside of them.

Differential Impedance

Differential impedance is established by the interaction of two traces and their reference plane(s). There are five basic differential impedance structures:

1. **Edge-coupled microstrip:** comprised of two adjacent traces on an outer layer with a single reference plane below it. The two traces are routed just like standard microstrip traces except that the space from trace to trace is significantly smaller than the distance of the traces to the plane.

2. **Edge-coupled embedded microstrip:** an edge-coupled microstrip line that has a dielectric over the top of it; solder mask will change a microstrip into an embedded microstrip line. The two traces are routed just like standard embedded microstrip traces except that the space from trace to trace is significantly smaller than the distance of the traces to the plane.
3. **Edge-coupled stripline:** a configuration with two adjacent traces on an internal layer, which is centered between a reference plane above and below it. The two traces are routed just like standard stripline traces except that the space from trace to trace is significantly smaller than the distance of the traces to the plane.
4. **Edge-coupled dual stripline or offset stripline:** an edge-coupled stripline, which is offset between the two reference planes; it is generally used when two adjacent signal layers are routed orthogonally and have reference planes outside of them. The two traces are routed just like standard dual stripline traces except that the space from trace to trace is significantly smaller than the distance of the traces to the plane.
5. **Broadside-coupled stripline:** a configuration with two differential lines on adjacent layers, one directly above the other.

Controlled Impedance Design Guidelines

- Standard impedance tolerance: $\pm 10\%$
- Tighter tolerances are available but need special care
- Broadside-coupled striplines should ideally be used on a core for better control of the trace-to-trace spacing of the layers. Try to avoid having prepreg between them. This is required to control the Z-axis alignment between the two signal layers
- Specify the design trace-to-trace spacing for correct modeling for differential structures
- Give the impedance lines a different aperture (usually 0.0001" [0.00254 mm] difference) than non-impedance traces. This makes it easier to adjust line widths if needed.
- Specify reference planes
- Reference plane must be contiguous under the entire transmission line

Minimizing Impedance Costs

Use the following guidelines to help to minimize impedance costs:

- Specify impedance only on layers that really require this
- Route all controlled impedance traces onto the same layer (typically on 4–6 layer PCBs)
- Specify a $\pm 10\%$ tolerance when possible
- Designate one target impedance value to be tested per layer
- Couple power/ground on adjacent layers when possible, and power/return layers need to be paired to achieve decoupling capacitance
- Allow for modifying the construction to meet overall tolerance (may require design authority approval to avoid unintended consequences)

Conformance Tests

100% Testing

All impedance coupons are 100% tested when specified by the customer. After comparisons against specified values, the measurements are electronically stored. Test coupon designs are defined in IPC-2221 and required when impedance is required.

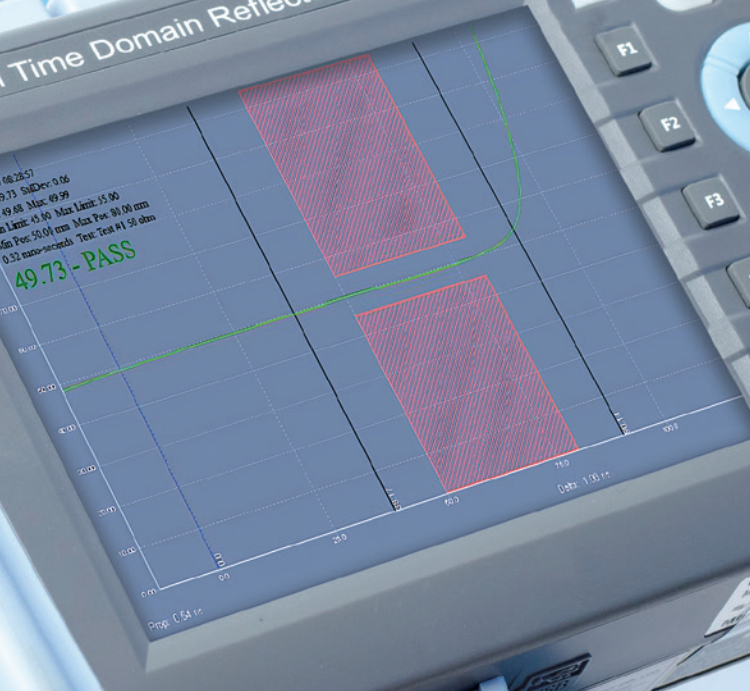
Serialization

Serialization is a traceability process for controlled impedance jobs. Impedance is typically measured with a TDR (Time Domain Reflectometer) by adding test coupons to the panel. Before the impedance has been tested, the coupon and PCB have a serial number marked on them by the fabricator.



Optical Time Domain Reflectometer

09-01-11 08:28:57
Avg: 49.73 StdDev: 0.08
Min: 49.68 Max: 49.99
Min Limit: 49.60 Max Limit: 49.90
Min Pos: 49.00 mm Max Pos: 50.00 mm
0.52 nano-seconds Test Test #1 20 ohm
49.73 - PASS



OTDR L5 VFL OPM

MENU REAL AVG AUTO

F1 ENTER F2 F3 F4 F5

A/B ESC SETUP

ON CHARGE

Optical Time Domain Reflectometer
Model: OTDR-1000
Max Wavelength: 1625 nm
Min Wavelength: 1270 nm
Polarization: PMD < 0.1 ps
Dynamic Range: 20 dB



Material Construction Stackups

A printed circuit board stackup refers to the material construction of the PCB, including core dielectrics, prepreg dielectrics, and copper sheets. Many factors need to be considered when developing a stackup, including overall thickness restrictions, circuit width and spacing, and impedance values required. The ASC design guidelines presented earlier, along with frequent engagement with your PCB fabricator, are always encouraged to select the optimum solution for any particular design.

Design success guidelines:

- Design multilayer boards with an even number of layers
- If specifying the dielectric thickness when, for example, it may be required for impedance reasons, the dimensions should be selected from available core or prepreg thicknesses
- Dielectric thicknesses made up of prepreg depend on the type or combination of suitable prepreg and achievable dimensions and tolerances
- It is beneficial to discuss special dielectric requirements during the design stage with your PCB vendor, if possible
- If the prepreg layer thickness/total dielectric thickness for the given layer is greater than 0.016" (0.406 mm) use filler/dummy cores (cores with no copper) to achieve the total layer thickness

Outer layer circuitry: Circuit area and distribution between the front and back of the board should be balanced as closely as possible. The addition of plating thickening to low pattern density areas on the external plane area may be considered.

Overall board thickness tolerance: As the overall thickness of a multilayer board increases, the thickness tolerance should also increase. A good rule is to specify a tolerance of $\pm 10\%$ of the overall thickness. Always indicate where the thickness measurement is to be taken, for example:

- Glass to glass
- Over gold contacts
- Over solder mask

Pure Build vs. Hybrid Build

The term “pure build” refers to a multilayer PCB material construction composed of the same type of material throughout the stackup, such as a construction entirely of FR-4, PTFE, or other high-frequency material.

However, a hybrid multilayer PCB uses materials with significantly different critical properties than those associated with a traditional “pure” multilayer PCB. The reasons for using a hybrid construction in multilayer PCB designs may be due to the necessity for RF or other specialty structures in the design in order to achieve higher reliability and/or other electrical performance requirements. Depending on the material set, these specialty structures can be more costly and often more difficult to process by the fabricator. On the other hand, using a hybrid of low-loss and FR-4 material can be more cost-effective than a pure low-loss package.

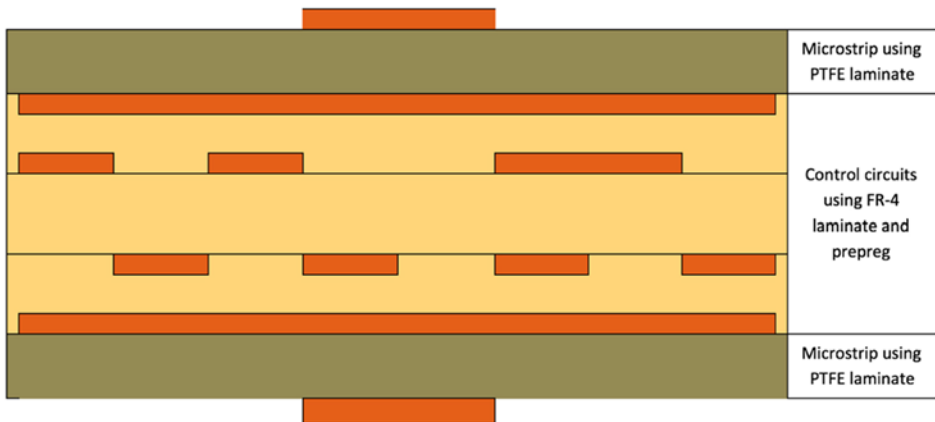


Figure 4.1: Typical microstrip construction.

Balanced Stackup Structure

Maintaining a balanced layout in relation to the Z-axis median of the board will assure minimum bow and twist. This balance includes the following:

- Dielectric thickness of layer
- Copper weight/thickness of layers
- Sequence/order of the plane vs. signal layers

A higher number of layers normally will mean an increased number of plane layers. Planes should be balanced around the Z-axis median line of the layup, and ideally located internal to the board. If accepted multilayer design rules are adhered to, boards will meet a maximum allowable bow and twist specification within 1.5% for through-hole and 0.75% for SMT technology per IPC-2221. A balanced construction is important with any PCB design, but is even more critical with a hybrid material set using mixed dielectrics.

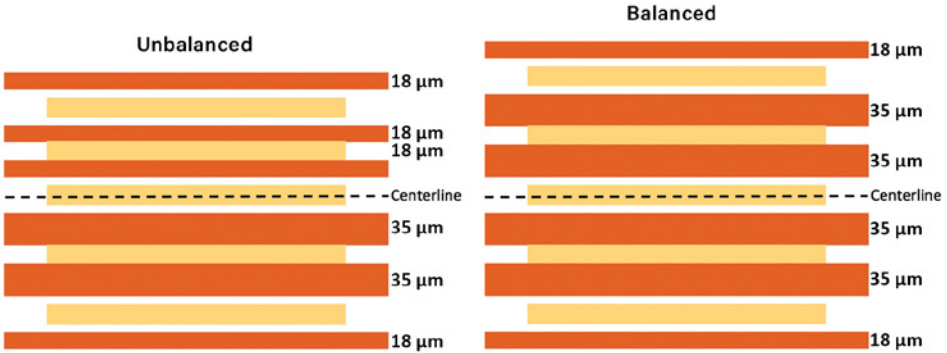


Figure 4.2: Unbalanced vs. balanced mixed-dielectric construction.

Signal Layers

A standard recommendation is that all pad-to-trace intersections be tear-dropped whenever the pad diameter minus the plated hole diameter is less than 0.020" (0.508 mm). Tear-dropping is a design technique that has a graduated fillet from the pad to the trace to increase interconnect reliability on the signal layers.

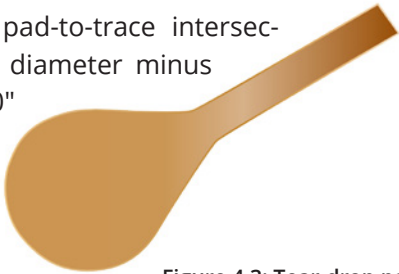
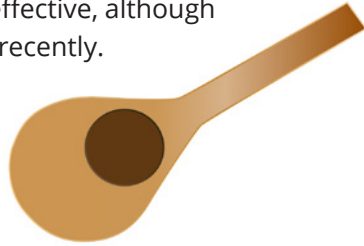


Figure 4.3: Tear-drop pad.

This process is designed to provide additional metal at the critical junction of a pad and a trace. When a panel is drilled and misregistration occurs, it has been shown that a long-term reliability issue can arise if the misregistration occurs at the junction of the pad and trace. Additionally, IPC-2221 requires a minimum amount of annular ring for the pad. Adding metal at this location helps ensure that an adequate connection is made and maintained. The tear-dropping process involves adding filleting to a secondary pad at the junction of an existing (primary) pad and a circuit trace. These secondary pads are sized 0.002" (0.050 mm) smaller than the primary pads, and the center is placed 0.003" (0.076 mm) away from the center of the primary pad. This tooling is conducted using IPC standards for tear-dropping

and has proven to be highly reliable and effective, although alternate solutions have been developed recently.

Figure 4.4: Tear-drop providing annular ring with off-center drilled hole.



Internal Power/Ground Layer Guidelines

- Provide a layer number and description on the artwork
- Clearance pads are recommended to be a minimum diameter of 0.020" (0.508 mm) larger than the nominal finished hole size. This clearance is dependent on the thickness of the copper layer. Consult your fabricator for specifics
- When placing thermal pads, size the outside diameter using this formula:

$$\text{ID} = \text{FHS} + 0.020" \text{ (0.508 mm)}$$

$$\text{Minimum: OD} = \text{ID} + 0.020" \text{ (0.508 mm)}$$

- Rotation of thermals and addition of spokes is customer dependent
- Preferred set: rotate each thermal 45° to the plane and add spokes that are 90° apart
- Spoke width:

$$\text{Standard: } 0.008" \text{--} 0.020" \text{ (0.203 mm--0.508 mm)}$$

$$\text{Minimum: } 0.006" \text{ (0.152 mm)}$$

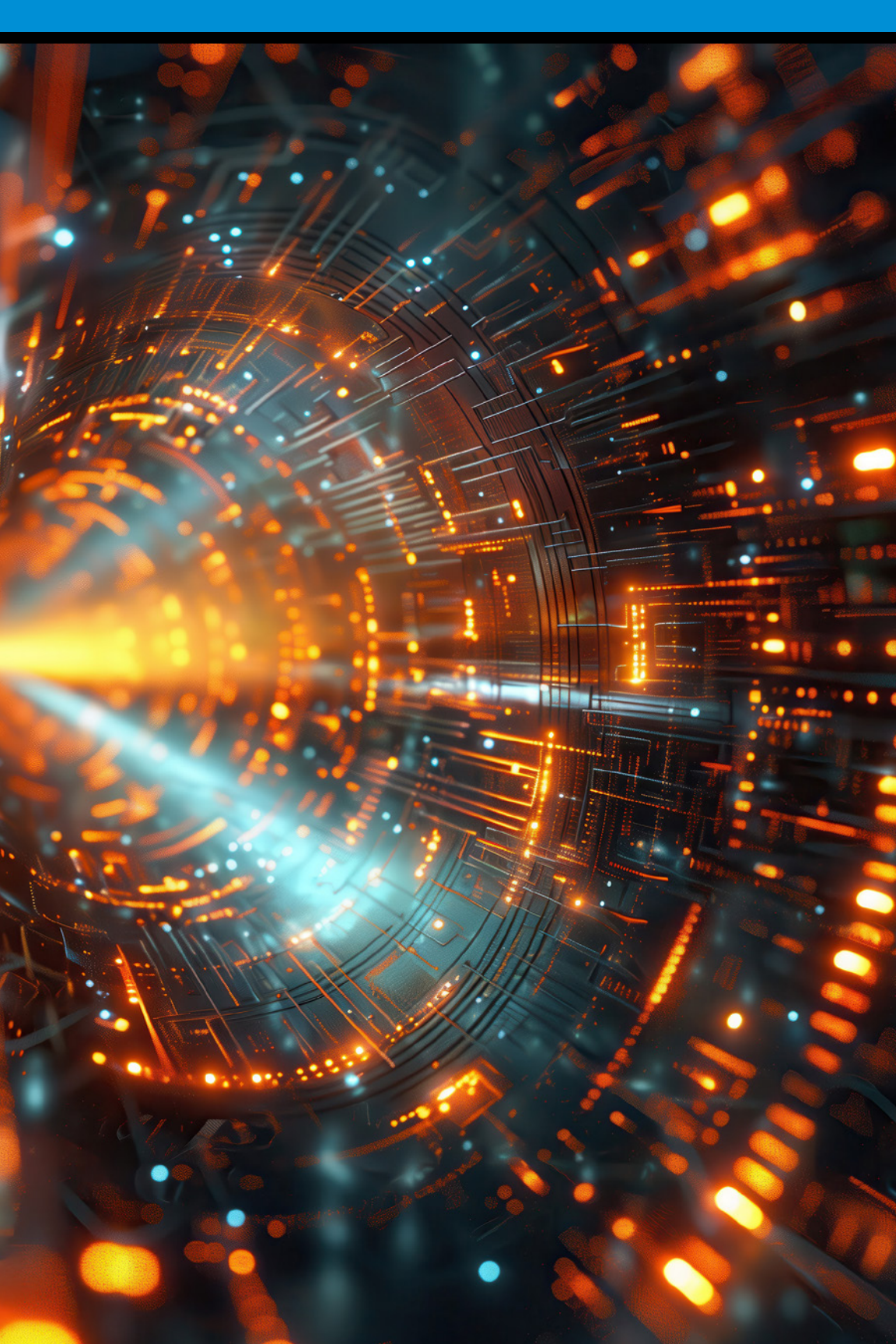
- To prevent exposed copper at board edge of routed panel, keep copper 0.010" (0.254 mm) away from the PCB's perimeter
- Minimum barrel of hole-to-copper spacing: 0.008" (0.203 mm)
- To prevent exposed copper at board edge of scored panel, keep copper at least 0.025" (0.635 mm) away from PCB's perimeter

Internal Signal Layer Guidelines

- Preferred trace/space: 0.004"/0.004" (0.102 mm)
- Minimum trace/space: 0.003"/0.003" (0.076 mm)
- Inner signal layers must have a positive polarity
- Clearly label layer number and description
- Minimum barrel of hole-to-copper spacing: 0.008" (0.203 mm)
- No thieving smaller than a 0.030" (0.762 mm) feature size is allowed
- Relieve all copper internal to part from route paths by at least 0.010" (0.254 mm)
- Provide thieving inside all open and breakaway areas, if permitted by the customer

Outer Layer Guidelines

- Preferred minimum trace/space: 0.004"/0.004" (0.102 mm)
- Minimum trace/space: 0.003"/0.003" (0.076 mm)
- Keep all pad edges at least 0.050" (1.27 mm) from gold features or they will be gold-plated
- On the solder side, leave a 0.200" x 0.400" (5.080 mm x 10.160 mm) area to add a date code box, ID, UL logo, cage code, etc.
- Layout of circuitry on the board has a major influence on the way the panel actually electroplates; try to avoid unbalanced copper areas on the outer layers. Solitary traces will over-plate, while isolated holes will over-plate to yield finished hole sizes under specifications
- To maximize plating distribution, allow the fabricator to add square thieving to low-density areas on the outer layers (such as breakaways or substantial unused spaces within the PCB)
- For boards with edge connectors, there must be at least 0.050" (1.270 mm) of solder mask between the solder mask clearance of the nearest via hole and the top of the edge connector area; if this spacing is violated, mask will be extended onto the edge connector until 0.050" (1.270 mm) is achieved



Via Structures

Standard Through-holes

Through-holes are holes drilled through the entire thickness of the board from the top layer to the bottom layer. They serve multiple purposes in PCB design and manufacturing, including:

- **Component mounting:** Through-hole technology (THT) components have leads that are inserted into these holes and soldered to pads on the opposite side. This method provides strong mechanical bonds and is often used for components subject to mechanical stress.
- **Electrical connections:** Through-holes can act as vias, providing electrical connections between different layers in a multilayer PCB. There are two main types of vias:
 - **Plated through-holes (PTH):** These holes have conductive plating, creating a reliable electrical path between layers.
 - **Non-plated through-holes:** These are not plated and are typically used for mounting purposes or as alignment holes.
- **Heat dissipation:** Through-holes can aid in heat dissipation by allowing heat to pass from one side of the board to the other.
- **Mechanical support:** Larger through-holes are sometimes used for mechanical support, such as screws or standoffs, to secure the PCB within an enclosure.

Design Considerations

- **Hole size:** The size of the hole must accommodate the component lead and allow for proper soldering. The hole diameter is typically slightly larger than the component lead diameter.
- **Annular ring:** This is the area of the pad surrounding the through-hole. Adequate annular ring size is crucial for a reliable solder joint.

- **Spacing:** Adequate spacing between through-holes and other PCB features is necessary to avoid electrical shorts and to comply with manufacturing capabilities.
- **Aspect ratio:** The aspect ratio is calculated by dividing the PCB thickness by the hole diameter. Typical aspect ratio guidelines are: standard aspect ratio $\geq 8:1$, high aspect ratio $\geq 10:1$ and ultra-high aspect ratio is $\geq 20:1$. Drilling machine capability and technology will have to be matched to the aspect ratio of any design as outlined in the section below.

Microvias

Buried and blind vias have been used in the industry for quite some time, but were limited by via size, officially defined in IPC-2226. One of the most important technological advancements that made HDI viable was the development of the microvia: a very small hole (typically 0.006" [0.152 mm] or smaller) that only connects certain layers either as "blind" or "buried" via holes. This represents a totally new way of making electrical connections between layers on a PCB.

Traditional PCB technology has utilized through-holes generated by mechanical drills. This limits the minimum size of via that can be fabricated. Mechanically drilled vias are typically drilled through the entire PCB, connecting the two outside layers with all of the internal layers. However, by also incorporating sequential lamination and controlled depth drilling, mechanically drilled vias can connect the same layers as microvias, only using a larger via size. The ability to strategically connect only certain pads on certain layers greatly

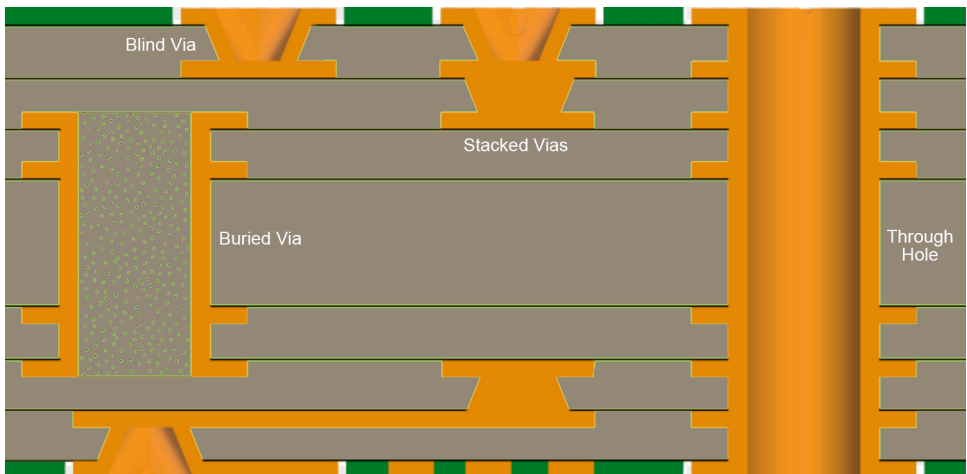


Figure 5.1: Microvias vs. through-hole vias.

reduces the real estate needed for a PCB design and allows a much greater density in a smaller footprint.

Types of Vias

- **Through via (through-hole):** This is the traditional via connecting the top layer all the way to the bottom layer
- **Blind via:** Used to connect one surface layer with at least one internal layer
- **Buried via:** Used to create connections of internal layers with no contact to the surface layers
- **Via-in-pad:** A type of blind via in which the via hole is drilled in a surface mount pad, eliminating the need to run a trace and via pad from each SMT pad. Typically done with microvias, but can be done with traditional blind vias as well
- **Filled vias:** Completely filling the via with either a non-conductive or conductive paste. Conductive fill is typically used for heat dissipation; non-conductive fill is used to closely match the thermal expansion of the substrate

Lamination Cycles

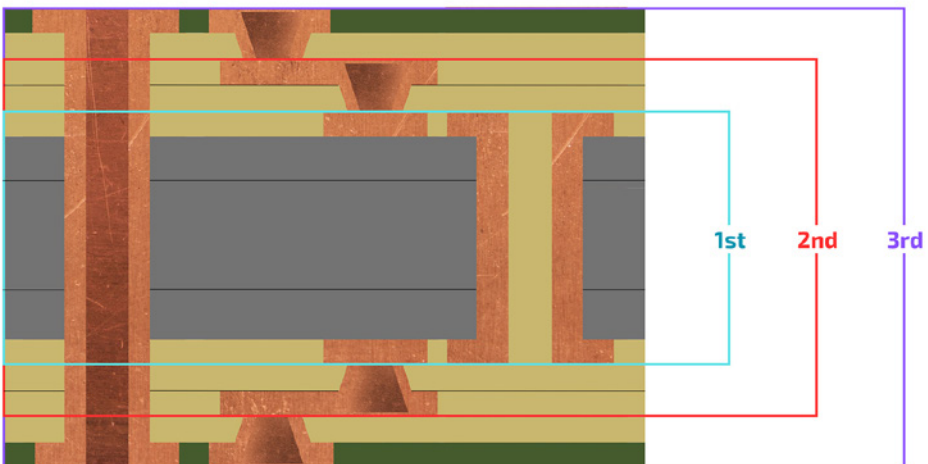


Figure 5.2: Sequential lamination with via structures.

Microvia Formation

Microvias can be formed through a number of methods, primarily plasma drilling or laser drilling.

- **Mechanical drilling:** Uses traditional drilling equipment to mechanically form holes, but typically limited to 0.006" (0.152 mm) diameter and dependent on the depth needed

- **Laser drilling:** Special drilling equipment that utilizes a laser to form the hole and can go down to 0.001" (0.025 mm) in diameter
- **Sequential lamination:** A process where the microvias are drilled all the way through a sub-panel of the layers that need to be connected by the via, which could require multiple lamination, plating, filling, and planarization operations

Stacked vs. Staggered Microvias

- **Stacked:** Microvias that are electrically connected and mechanically stacked vertically on top of each other through various layers of the PCB
- **Staggered:** Microvias that are electrically connected and offset to one another through various layers of the PCB

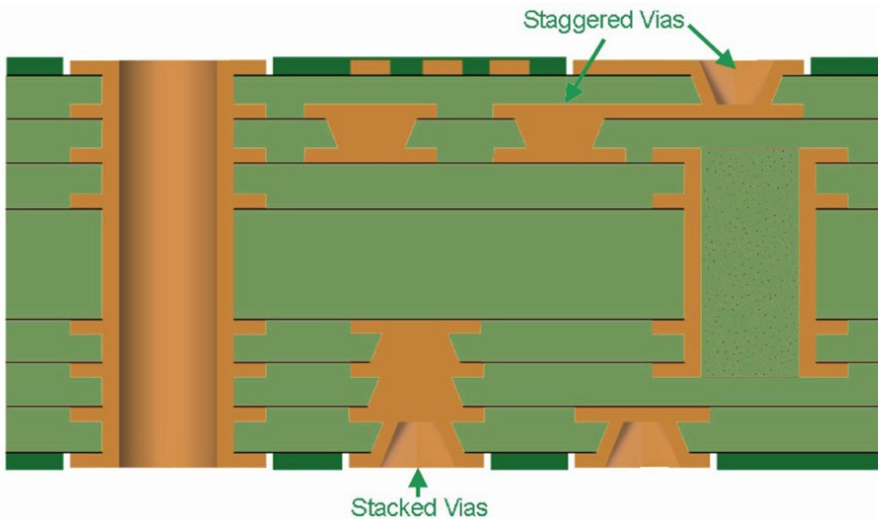


Figure 5.3: Staggered and stacked microvias.

Via-in-Pad Microvias

The via-in-pad production process allows you to place vias in the surface of the flat lands on your PCB by plating the via, filling it with one of the various fill types, capping it, and, finally, plating over it. Via-in-pad is typically a 10- to 12-step process that requires specialized equipment and skilled technicians.

Via-in-pad is often an optimum choice for HDI PCBs because it can simplify thermal management, reduce space requirements, and provide one of the shortest paths to bypass capacitors for high-frequency designs.

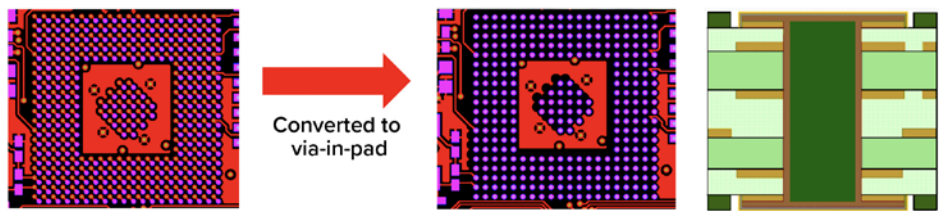
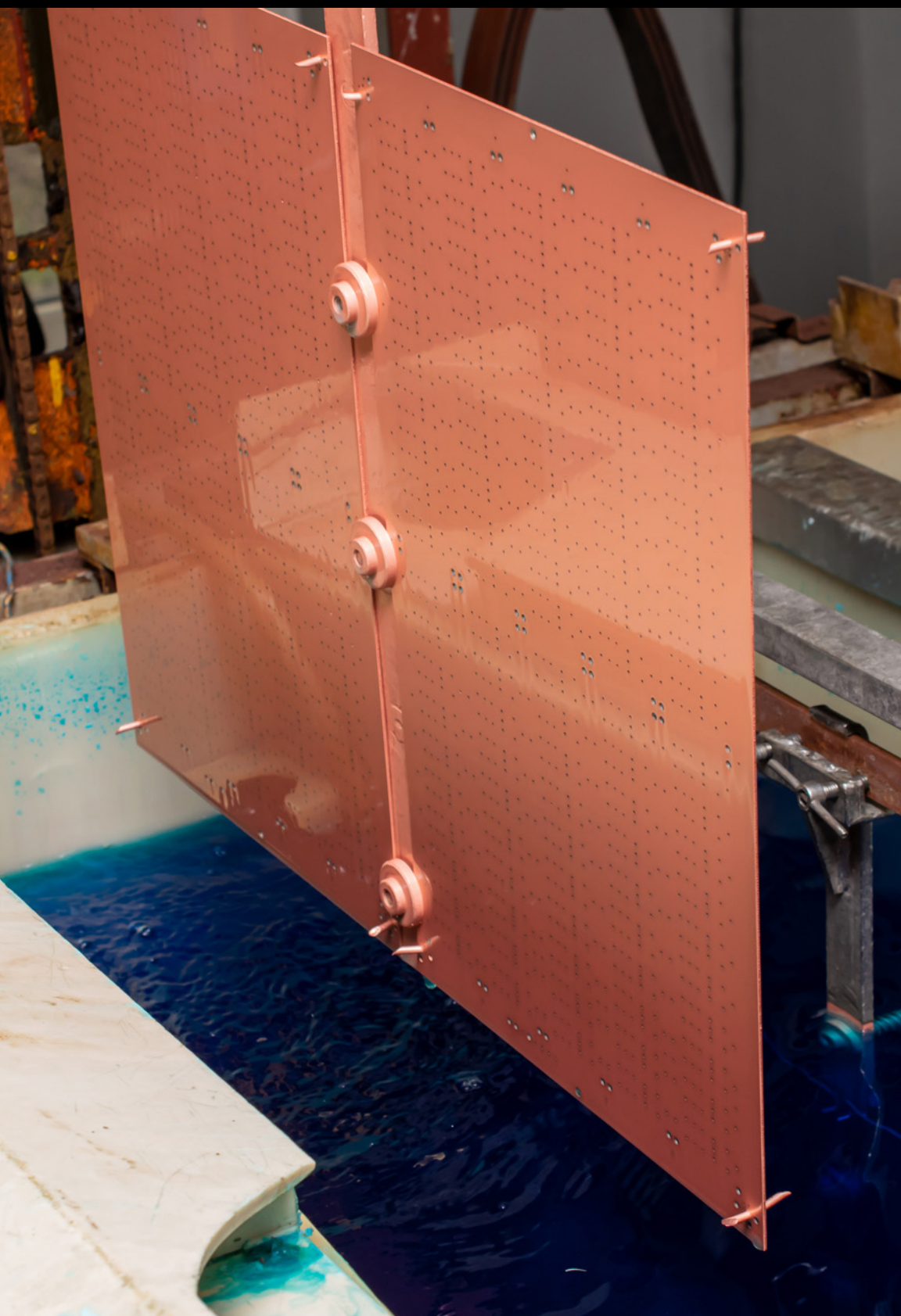


Figure 5.4: Comparison of typical layout to via-in-pad, showing savings in area.



Plating Methods

Electroless Plating

Electroless copper plating is a specific application of electroless plating tailored for depositing an ultra-thin layer of copper onto the surface and through-holes in PCB panels. This process is integral to the production of PCBs, providing the initial conductive layer of copper from the trace surfaces through the drilled holes.

Electroplating

There are two methods of plating copper on PCBs: pattern plating and panel plating. The panel plating method eliminates most of the copper plating distribution issues, but, because it adds copper thickness to the base layer, it makes maintaining fine line definition and consistency difficult. Base copper is often measured in ounces of copper per square foot of surface area (½-ounce copper is ~0.7 mils thick; 1-ounce is ~1.4 mils thick).

Pattern Plate

This standard process has major advantages in that only the base copper is required to be etched. This process yields finer, better-defined lines (traces). One possible disadvantage is the variations in trace height due to distribution issues during electroplating.

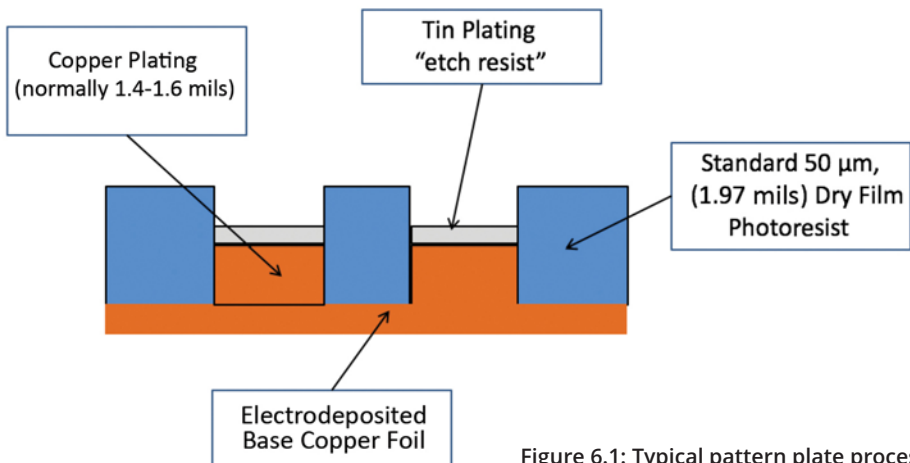


Figure 6.1: Typical pattern plate process.

Panel Plate

This plating fabrication method eliminates most of the copper plating distribution issues but now the excess surface copper (after the circuit pattern is defined) must be etched along with the base foil. This makes maintaining fine-line definition and consistency difficult.

The allowance for etching of plated designs is shown in Table 6.1.

Table 6.1: Allowance for etching of plated designs.

Copper Weight	Typical Reduction in Line Width From Nominal	Minimum Design Line Width
0.25 ounce	0.0005"	0.003"
0.5 ounce	0.0007"	0.004"
1.0 ounce	0.0014"	0.005"
2.0 ounce	0.0028"	0.008"

It is important to mention that the guidance shown here reflects raw copper foil weights and not any additional design features, such as via fill or sequential blind laminations, that will add copper thickness to the foil.

Design Compensation

Because of the physics of producing printed circuits, the PCB fabricator first needs to plate the circuit pattern (pattern plate), followed by a chemical etching (subtractive) process to define the circuit line width and space. To accomplish this, the initial circuit definition will need to be compensated for line width loss due to etching to ensure that the final circuit will meet the design criteria.

Surface-to-hole Plating Ratio

Regardless of how much copper a fabricator begins with, additional copper on the surface will be plated (pattern plating) as the plated through-holes are metallized. The key here is that this is not a 1:1 ratio; more copper will always be deposited on the surface than in the holes due to a higher plating current density on the surface. A working guide would be to calculate that 1.2 to 1.4 times the average minimum hole thickness will be deposited on the surface. Using the industry standard requirement of 1.0 mils in the hole, the surface would receive 1.2 to 1.4 mils of plated copper on the surface, plus any base copper thickness. Additional guidance can be found in IPC-6012.

Impact of Sequential Laminations and Via Fill

Sequential lamination (sub-assembly processing) is used to manufacture

buried/blind vias, via fill, microvias, etc. These technologies further compound the final surface copper thickness considerations discussed above, as each process requires additional plating and etching processes. Designers need to fully understand the impact of additional plating processes on the final thickness of copper traces and surfaces.

Edge Plating

Edge plating is the encapsulation of the edges of printed circuit boards to improve EMI shielding of higher-frequency designs and chassis ground in systems. The requirement for edge plating is being implemented for single-axis and multiple-axis edges of the circuit board, including all four edges.

Edge plating is created when a rout path is implemented prior to the metalization of the circuitry features of the printed circuit board and is sometimes

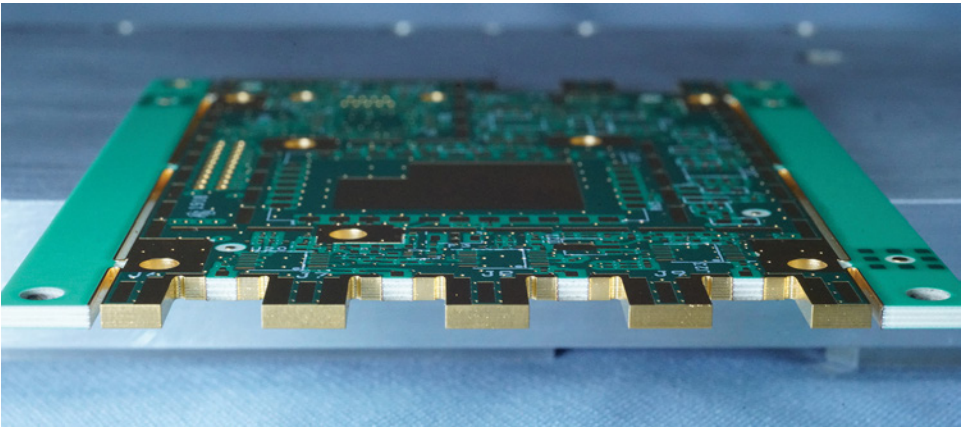


Figure 6.2: Example of edge plating.

referred to as “plated rout.” The design requirements for implementing this technology are dependent on the requirement for the number of edges of each board, the size of the board, and whether the boards will be delivered in a multi-up array.

The stability of the material used for manufacturing the board will also play a part in the development of the routing requirements. Materials where the reinforcement is not the standard glass-reinforced FR-4 type may require an alternative routing design to improve the structure of the parts inside the panel that is used to carry the parts during manufacturing. The parts are processed still intact inside the carrier panel, also called an array, used for manufacturing and assembly, requiring the development of tabs, in some cases, for edge plating. For a single edge requiring plating, tabs may not be required for stability.

Design Rules for Plated Edges

It is required that the plating not only encapsulate the edge of the panel, but also wrap around to the surface of the panel as shown in Figure 6.3. The minimum distance the plating is required to wrap around to the surface is 0.015" (0.381 mm). The wrap-around is required for internal processing and adhesion of the plating to the edge. Also detailed is the minimum distance a feature can be placed to the wrap-around plating (0.010" [0.254 mm]) as well as the distance required for an adjacent routed feature (0.100" [2.540 mm]).

It is typical that the plating be continuous along the entire edge of the circuit, but interruptions in the plating can be designed with two methods. By placing a tab at the interrup-

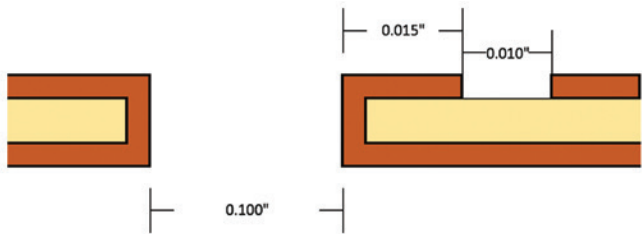


Figure 6.3: Example of wrap-around proper edge plating.

tion, and subsequently removing it, the tab prevents plating in this area and is removed at the final routing stages. The subsequent removal of the plating will leave a small tab or an indentation in the circuit. The indentation is created when the router bit contacts the edge where the plating is removed and advances into the circuit to completely remove the plating. When the interruption of the plating is a minimum of 0.200" (5.080 mm), the use of a tab to prevent continuous plating can be used. When less than 0.200" (5.080 mm), the edge will be plated entirely across and subsequently removed by the routing process described.

Edge plating generally involves connecting one or more of a board's internal planes. These planes extend to the edge of the board and are electrically connected by the plating process. Those layers are typically fabricated with a polarity border (ref. 0.050" [1.270 mm]) to prevent exposed copper when they are routed and removed from a panel.

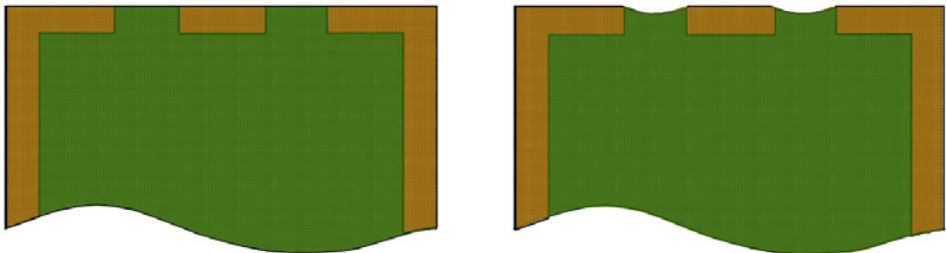


Figure 6.4: Panels with polarity borders.

Breakaway Tabs for Arrays

Breakaway tabs, also known as *perforated tabs* or *mouse bites*, are common features on printed circuit board arrays. They are used to hold multiple PCBs together during manufacturing and assembly processes, allowing for efficient fabrication and handling. Figure 6.5 shows how perforated tabs are implemented for a circuit board that is manufactured in a multiple-up array. The requirement is that the tabs being used to hold the board in the array will be broken off after the assembly operation (also called singulation). The image on the left shows a perforated tab while the one on the right shows a solid tab that is placed at the plated rout process. In both cases, the internal and external layer artwork around the tabs must be void of metal. This will prevent the metal from being exposed in the singulation process. When a solid tab is used, the tab is removed at the final fabrication stage using a router (the same router that removes standard parts from the carrier panels). Removal of the tab will leave a small protrusion along the edge of the part.

The size and shape of breakaway tabs can vary depending on the specific requirements of the PCB design and manufacturing processes. Common shapes include rectangular, circular, or hexagonal tabs. The size of the tab should provide sufficient strength to hold the PCBs together during fabrication and assembly, but also be easy to break or remove when necessary. Breakaway tabs are typically placed along the edges of the PCB array and can be located either along the outer edges or within the array, depending on the design requirements. Tab design factors to consider include the PCB manufacturing process, assembly requirements, materials being used to construct the circuits, and thickness of the PCB. Tab length can range from 0.100" (2.540 mm) to 0.200" (5.080 mm) (depending on the above considerations). Very thin and/or flexible PCBs may require tabs <0.100" (2.540 mm) in length. Conservatively, the tabs are placed every two inches along the plated edges. Additional guidance can be found in IPC-2221 and IPC-2222.

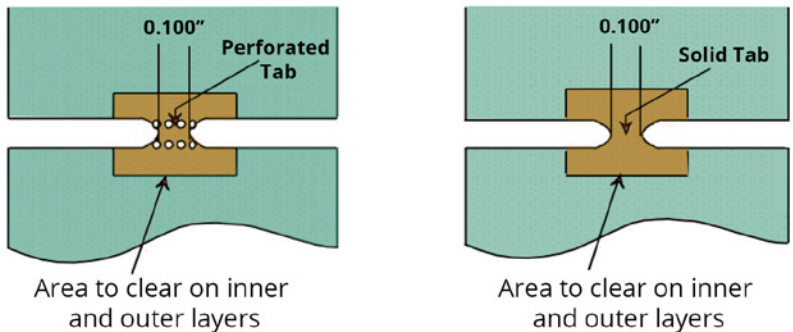
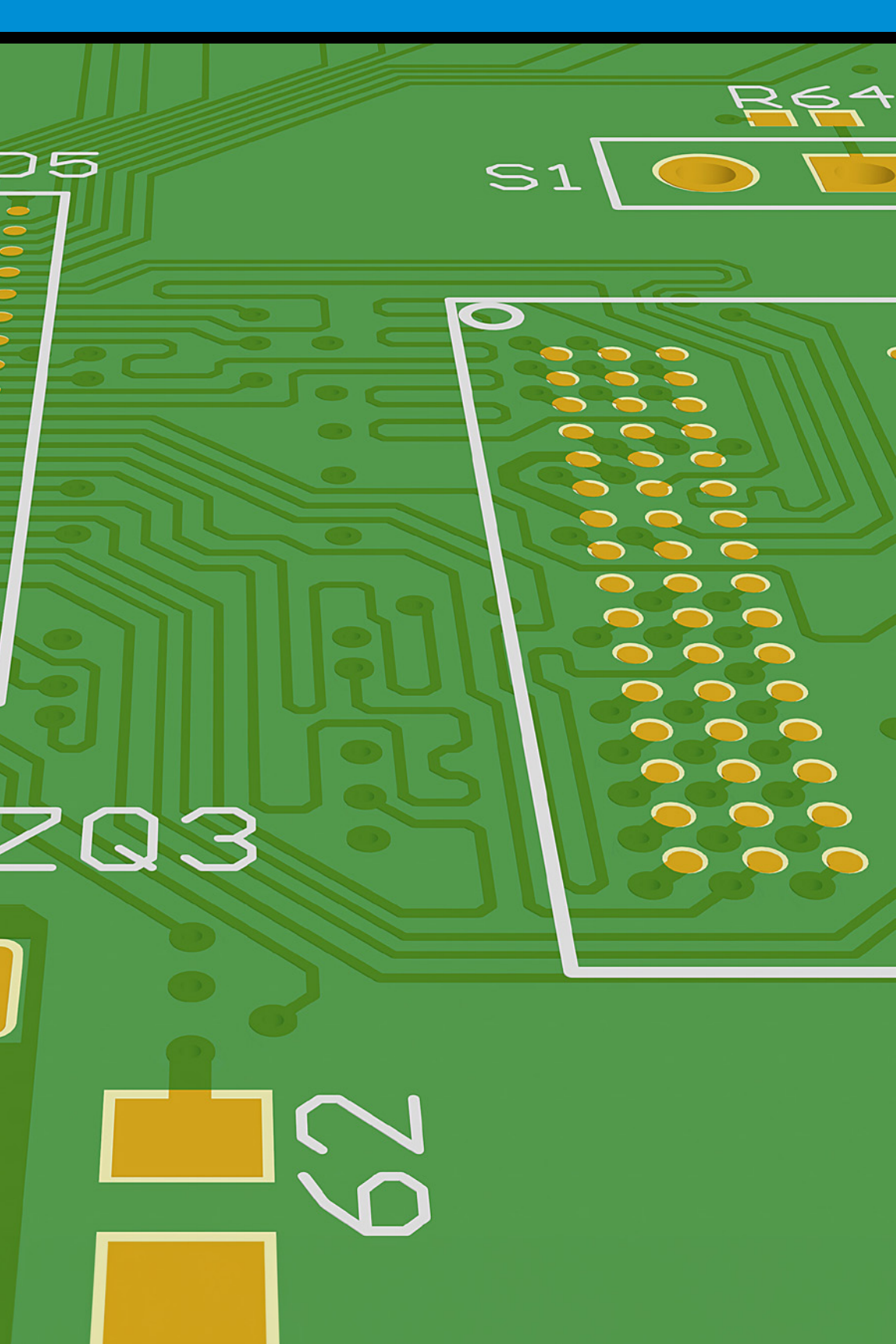


Figure 6.5: Perforated vs. solid tab breakaway design.



Solder Mask and Legend

Solder mask is a protective coating that shields selected areas of a PCB from oxidation, handling, and unwanted solder during assembly. Green is the most common solder mask color.

Solder mask application typically allows a 0.003" (0.076 mm) web to be placed between pads and other features on the PCB, provided the minimum spacing between these features is 0.007" (0.178 mm) by design. Not all colors can maintain this web size. It is recommended that requirements and capabilities be discussed with the fabricator.

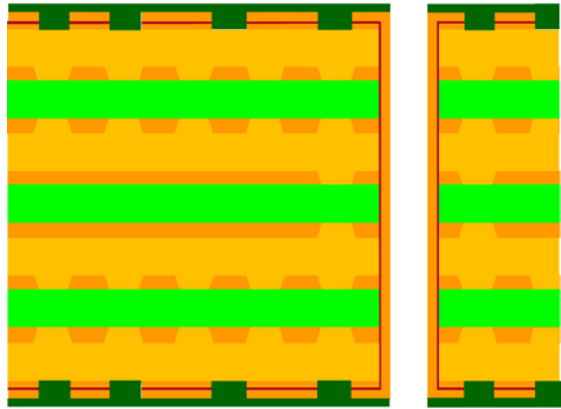


Figure 7.1: Illustration of solder mask on outer layers (dark green).

To ensure there is no solder mask on any pad in an SMD array, the minimum solder mask clearance for a surface mount pad is 0.002" (0.051 mm) per side. This is not applicable for panel sizes over 18" x 24" (45.72 cm x 60.96 cm). As space permits, a clearance of 0.0025"/0.064 mm per side is preferred. Allow 0.030" (0.762 mm) per side solder mask clearance for score lines. To prevent solder mask from going into and/or plugging a hole, the mask clearance should be 0.010" (0.254 mm) (0.005" [0.127 mm] per side) larger than the pad size on both sides of the board.

Most PCB fabricators provide a variety of solder masks to satisfy the needs of all customers. Standard imaging equipment and processes are fine for standard technology, but advanced technology with tighter registration tolerances has driven the implementation of laser direct imaging (LDI) solder masks (laser-defined solder mask pattern instead of screening).

Tenting of Via Holes with Solder Mask

Via hole "capping/tenting" is available through the via cap process. On a board coated with LPI (liquid photoimageable) mask, the vias can be screened with an epoxy or acrylic solder mask material, creating a cap over the hole. Artwork

modifications necessary for processing are performed as part of the initial tooling. A solder mask design file must be provided by the customer, which includes only those vias that are to be capped. The customer needs to provide master pad solder mask and via fills, i.e., solder mask and via pads that are the same size as the outer layer pads.

Via Capping Design Constraints

The maximum finished hole size for via capping is 0.020" (0.508 mm) diameter (preferred drill diameter 0.021" (0.533 mm). Generally, the non-test vias are capped on the top side of the board. Thermal (epoxy) via caps will have a raised surface of approximately 0.0017" ± 0.0004" (0.0043 mm ± 0.001 mm) above the outer layer copper pad, provided the via hole being capped is not plated shut or plugged with solder from HASL. In this case, due to trapped air between the via cap and plug, the via cap may “dome” during the cure process, thus creating a height of 0.0035" (0.089 mm). A UV (acrylic) via cap will have a raised surface of approximately 0.0019" ± 0.0004" (0.005 mm ± 0.001 mm) above the outer layer copper pads. Unlike thermal (epoxy), UV (acrylic) is not influenced by plated or HASL (solder) plugged holes; therefore, the via cap height will remain constant.

Note: *Via cap height thickness measurements may include HASL (solder) and/or permanent solder mask thickness.*

For product that receives either immersion silver or tin finish, UV via cap material must be used and applied after the surface finish.

Legend Marking Design Guidelines

Modern legend technology uses an inkjet printer for black or white legend; traditional technology is to screen it on, but with new printing technology coming out regularly, understanding your fabricator’s capability is critical. To ensure all letters, numbers, and figures are legible on the finished board, height-to-width ratios need to be considered and discussed, e.g. character line widths should be greater than 0.006" (0.152 mm) and at least 0.030" (0.762 mm) high.

Table 7.1: Legend marking design guidelines.

Specification	Standard Production	Advanced or Prototype
Type	Ink Jet Printer	UV / Thermal
Colors	White, Black	All
Smallest Line Width	0.003" / 75 μm	0.010" / 254 μm

Space letters at least 0.008" (0.203 mm) apart so they don't bleed together. No legend nomenclature should overlap a copper pad or plane area, which is especially important for surface mount pads and fiducials. White is the standard legend ink color. Use the fabrication print notes to specify special features to be screened onto the board, and non-standard colors. All legend should be kept at a minimum 0.003" (0.076 mm) away from solder mask clearance.

Nomenclature

Letter size minimums:

- Screened nomenclature: 0.006" (0.152 mm) line width
- Photoimageable: 0.004" (0.102 mm)

Recommended letter sizes:

Line Width	Height and Width
0.004"	0.020"
0.005"	0.025"
0.006"	0.030"
0.008"	0.035"
0.010"	0.040"

***Note:** Nomenclature placed over parallel or heavy copper traces may have poor legibility.*

Colors:

- White (epoxy) preferred
- Yellow (epoxy)
- Black and white (LPI ink) is also available
- Additional colors are available

Considerations

Nomenclature over solder (HASL) will have poor adherence. Nomenclature placed over bare copper before HASL and immersion silver/tin will have an apparent copper "halo" after the HASL or immersion silver/tin. The artwork will need to be modified if this is not acceptable to the customer.

FINAL FINISHES



ENIG

electroless nickel,
immersion palladium,
immersion gold

ENIG

electroless nickel,
immersion gold

ENEPIG

electroless nickel,
electroless palladium,
immersion gold

ImmAg

immersion silver

HASL

hot air solder
leveling

- SnPb 63/37 tin/lead
- LF (Pb-free)

Sulfamate Nickel

hard or soft gold
electrolytic
nickel/gold

ImmSn

immersion tin

OSP

organic solderability
preservative

Final Finishes and Design Guidelines

Final finishes provide a surface for the component assembler to either solder, wire bond, or conductively attach a component terminal or lead to a pad, hole, or area of a PCB. The other use for a final finish is to provide a known contact resistance and life cycle for connectors, keys, or switches, and also provides protection against oxidation for the exposed copper. The primary purpose of a final finish is to create electrical and thermal continuity with a surface of the PCB.

There are several final finishes in use in the industry today, including:

- ENIG (electroless nickel/immersion gold)
- ENEPIG (electroless nickel/electroless palladium/immersion gold)
- ENIPIG (electroless nickel/immersion palladium/immersion gold)
- Electrolytic Ni/Au (electrolytic sulfamate nickel/hard or soft gold)
- ImmAg (immersion silver with anti-tarnish)
- ImmSn (immersion tin)
- HASL (hot air solder leveling)
 - ◊ SnPb (63/37 tin/lead)
 - ◊ LF (lead- or Pb-free)
- OSP (organic solderability preservative)

Final finishes are primarily application driven, so there are several considerations that should be part of any decision to choose a final finish:

- Pb tolerant or Pb-free (LF) process
- Shelf life
- Flatness
- Coplanarity
- Lead or ball pitch
- Wire bondability
- Lead insertion
- Solder joint integrity
- Corrosion resistance
- Potential problems
- Cost

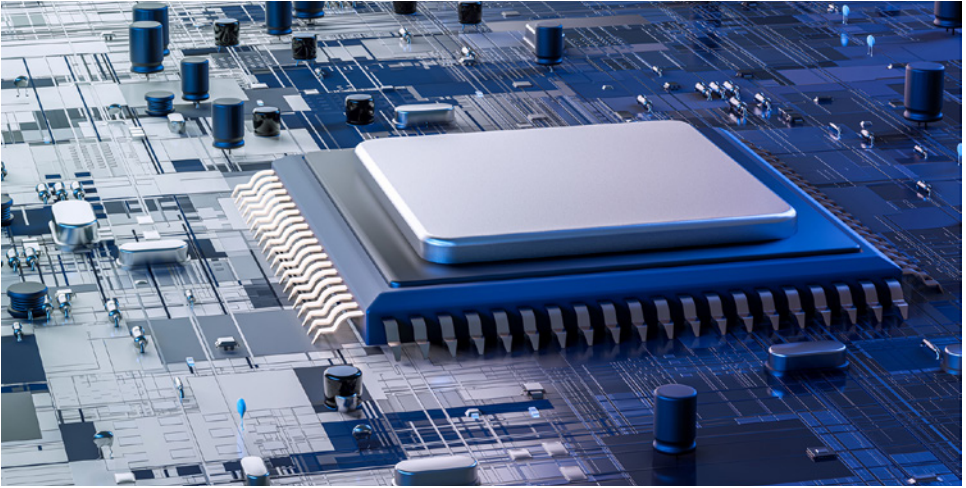


Figure 8.1: ImmAg (immersion silver) PCB after assembly.

Pb-free finishes are considered RoHS compliant ($< 0.1\%$ BW of finish, for Pb, Hg or Cd) with the single exception of tin/lead HASL. The RoHS-compliant finishes include the following:

- ENIG
- ENIPIG
- ENEPIG
- ImmAg
- ImmSn
- Electrolytic nickel/gold
- LF HASL
- OSP

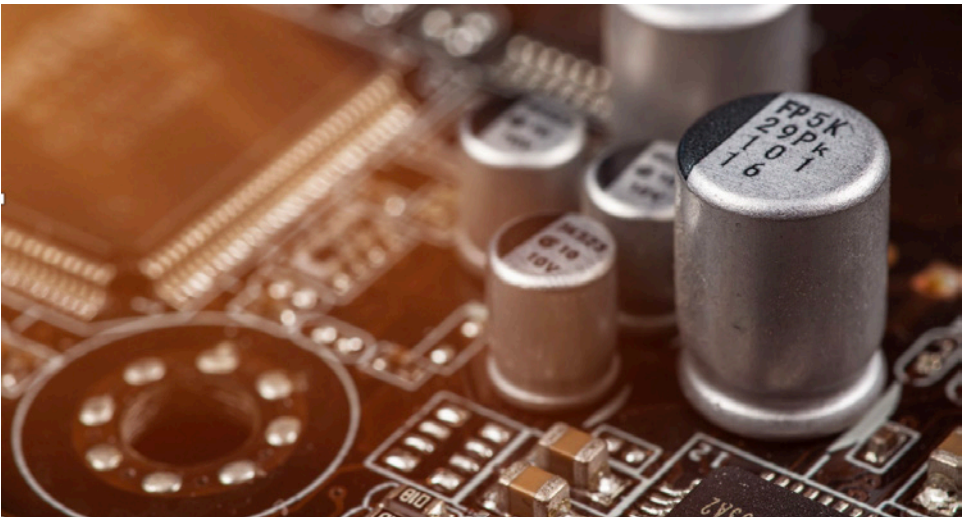


Figure 8.2: ENIPIG (electroless nickel/immersion palladium/immersion gold) PCB after assembly.

Pb-free PCBs require that the standard HASL surface finish cannot be used. There is still a significant amount of discussion on what the long-term surface finishes will be. Currently, the immersion silver and OSP surface finishes are the most prevalently specified surface finishes for solderable PCBs. Immersion tin is the prevalent surface finish for press-fit backplanes. Please contact your PCB fabricator for current information on where industry specifications are heading.

ENIG (Electroless Nickel/Immersion Gold)

ENIG is a two-layer metallic coating of 1.97 μm (0.05 μm) minimum, 2–5 μm (0.05 to 0.012 μm) immersion deposited Au over 118.1–236.2 μm (3 to 6 μm) electroless deposited nickel, per IPC-4552 nominal pad size of 0.060" x 0.060" (1.5 mm x 1.5 mm). ENIG is typically used for surface/fine-pitch devices. The benefits of ENIG are a long shelf-life, environmental resistance, as well as low contact resistance.

This process plates a thin coat of nickel covered by a thin layer of gold. The gold provides a very good solderable surface. When components are soldered onto these pads, the gold diffuses into the solder joint. The gold layer is very thin, so it won't reduce the solder joint strength. This process is generally not utilized for high reliability, long lifetime, or high vibration applications.

Aluminum wire bonding (not gold)

- Expensive
- Cannot be reworked
- Lossy RF (phosphorus nickel)
- Black pad potential

ENEPIG (Electroless Nickel/Electroless Palladium/Immersion Gold)

ENEPIG is a three-layer metallic coating of 1.97 μm (0.05 μm) minimum, 2–5 μm (0.05–0.012 μm) typical, immersion-deposited Au over 2–12 μm (0.05–0.30 μm) electroless-deposited Pd over 118.1–236.2 μm (3–6 μm) electroless-deposited nickel.¹

This process deposits a thin coat of nickel covered by a thin layer of palladium, then gold. The gold provides a very good solderable surface. When components are soldered onto these pads, the palladium and gold diffuse into the solder joint. The gold layer is very thin so it won't reduce the solder joint strength. This process is generally not utilized for high reliability, long lifetime, or high vibration applications. This is used for wire-bonding applications.

ENEPIG is typically used for flat surface/fine-pitch devices. The benefits of ENIPIG are a long shelf life, environmental resistance, low contact resistance, and excellent solder joint with SAC solders.

Gold Wire Bondable

- Expensive
- Cannot be reworked
- Lossy RF (phosphorus nickel)
- Palladium intermetallic bond to Pb in Sn/Pb is not ideal

ENIPIG (Electroless Nickel Immersion Palladium Immersion Gold)

ENIPIG is also a three-layer metallic coating surface finish that is a variation of the more common ENEPIG finish, but with the replacement of the electroless palladium layer with an immersion palladium layer. As with all gold layers, it offers excellent solderability and resistance to oxidation, ensuring reliable solder joints during PCB assembly.

ENIPIG finishes are chosen for applications where high reliability, excellent solderability, and resistance to surface corrosion are essential, such as in aerospace, automotive, medical devices, and telecommunications industries.

Electrolytic Nickel/Gold (Electrolytic Nickel and Hard or Soft Gold Plating)

Electrolytic nickel/gold is a two-layer metallic plating of a varied thickness of either hard or soft electroplated gold over a varied thickness of "low stress" electroplated sulfamate nickel.¹

Electrolytic nickel/gold has the best shelf life and environmental resistance, as well as a low contact resistance. Selective finishes are possible with electrolytic nickel/gold (<15 µin for soldering, >30 µin for contacts, and wire-bondable soft gold).

Soft Wire Bondable Gold

- Gold wire bondable
- Most expensive (thicker gold = \$\$\$)
- Selective finishes more expensive (filled vias required in some cases)
- Cannot be reworked
- Moderately lossy RF (nickel)

IAg or ImmAg (immersion silver)

ImmAg is a single-layer metallic coating of 5 μin (0.12 μm) minimum, 5–20 μin (0.12–0.50 μm) typical, of immersion-deposited silver.² This process plates a thin layer of silver directly on top of the copper surface. As with the other immersion surface finishes, the finished product produces a very flat surface, which is ideal for fine pitch SMD arrays. This surface finish has the ability to maintain high solderability after multiple heat cycles. This can also be used as an aluminum wire-bondable surface. It is compatible with no-clean assembly processes. This is becoming popular as a HASL replacement for Pb-free soldering applications. This surface finish will yield a dull tarnished looking surface. There is significant industry data showing that the dullness does not affect solderability or reliability.

ImmAg Attributes

- Inexpensive. It is typically used for flat surface/fine pitch devices and is aluminum wire bondable
- Good RF performance if the following guidelines are followed:
 - ◊ Packaging and storage are critical
 - ◊ Careful handling required
 - ◊ High insertion friction—not well suited for press fit pins

ImmSn (Immersion tin)

Immersion tin is a single-layer metallic coating of 15.7 μin –23.7 μin (0.4–0.6 μm) of immersion deposited Sn.³ This process coats a thin layer of tin directly on top of the copper surface. The tin produces an extremely flat surface for mounting of surface mount components with ultra-fine-pitch devices. This also results in a thicker uniform surface that provides lubrication for press-fit pins. ImmSn is inexpensive and typically used for flat surface. ImmSn is okay for fine-pitch devices with solder mask dams greater than 5 mils (127 μm), but is not the top choice for press-fit pin insertion for the following reasons:

- Careful handling required
- Poor environmental resistance of intermetallic
- Not great for multiple reflow cycles

It also must be noted that there is a higher risk of tin whiskers when using high-purity tin material.

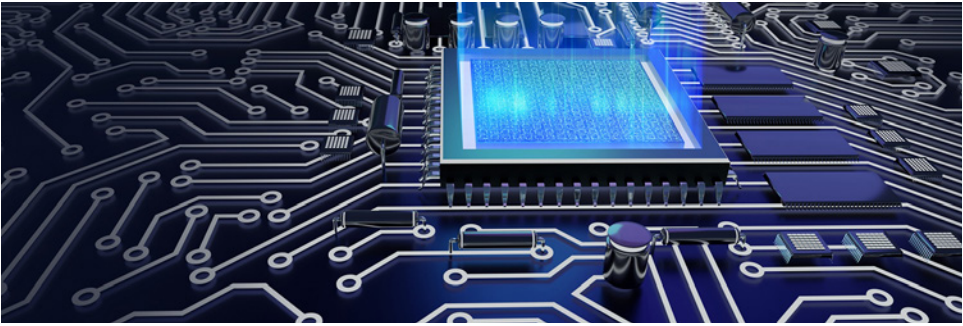


Figure 8.3: IAg or ImmAg (immersion silver) example.

HASL (Hot Air Solder Leveling)

HASL is available in two versions: a tin/lead alloy and a Pb-free option. HASL is a variable-thickness metal alloy coating, typically 25–2000 μm (0.65–50 μm) thick, designed to create a solderable surface and to maintain solderability over an extended period. Using specialized equipment, panels with exposed copper (pads and/or traces) are inserted into a vat of molten solder. A series of compressed hot-air knives then remove excess solder and “level” the surface before the solder solidifies and cools. HASL is the most prevalent and most solderable finish. Boards that have been HASL leveled will have a bright, silvery pad coloration.

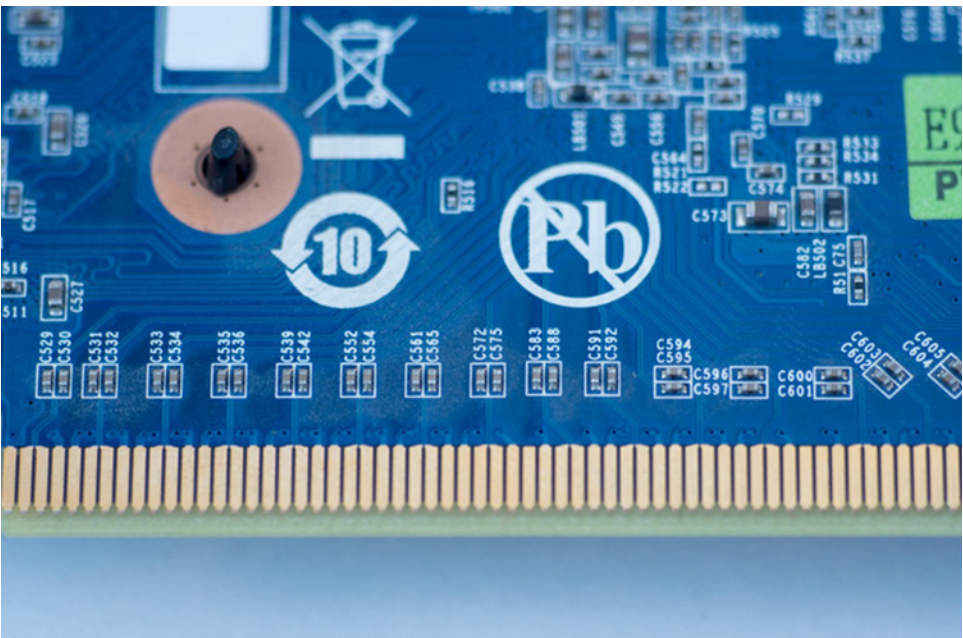


Figure 8.4: Pb-free HASL PCB after assembly.

Solder thickness and uniformity on SMD pads is a function of pad size, pad orientation during processing, and HASL equipment. As pad size increases, thickness and uniformity decrease. This is caused by the high surface tension of solder and the airflow over pads that remove excess solder. Small pads (0.025" [0.635 mm] pitch SMD) will have a more uniform solder coating than large pads (0.050" x 0.050" [1.270 mm x 1.270 mm]). Large pads may be partially covered with a mound of solder while the rest of the pad is thin eutectic solder. Small pads may have the same condition depending on pad orientation during processing.

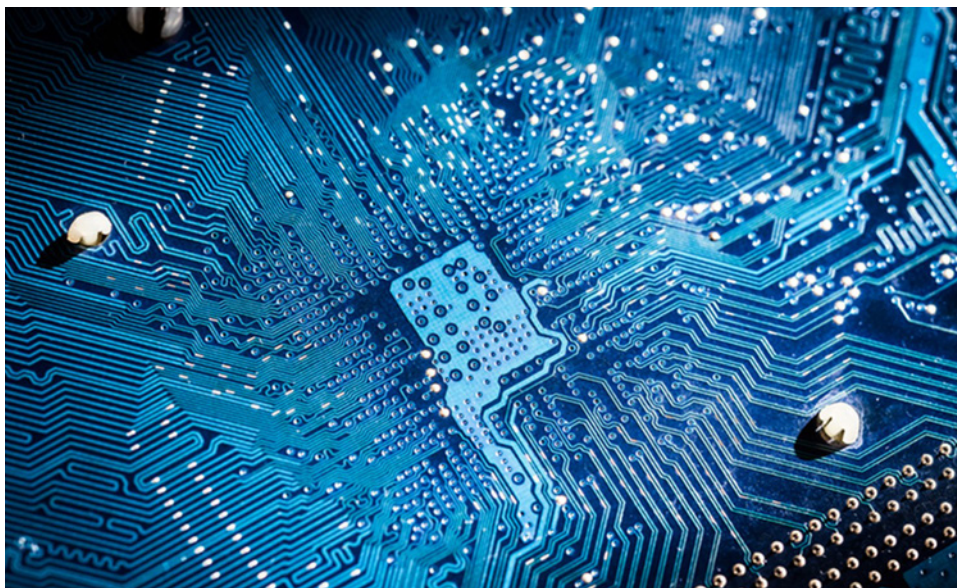


Figure 8.5: High-density PCB example.

Pad orientation also contributes to solder thickness and uniformity. Fine-pitch SMD pads perpendicular to the direction of processing will have a uniform solder coating covering the entire pad. Fine-pitch SMD pads that are parallel to the direction of processing will have the leading half reduced to the thin eutectic solder with a mound of solder on the trailing half (eutectic solder is the optimum metal alloy to maximize solderability). Angled processing exposes both pad directions to similar air blasts resulting in more consistent pad thickness for similar pad sizes. Processing at a 45-degree angle minimizes the variation in pad orientation. HASL is not used for fine-pitch components because it increases fabrication and assembly rework for boards that have fine-pitch SMT devices with lead centers below 0.020" (0.508 mm) and presents coplanarity issues.

Pros and Cons of HASL

Pros

- Economical and widely available
- Excellent solderability
- Can be reworked at the fabricator
- Presents a very flat solder deposit

Cons

- Limited assembly reflow cycles, as the HASL process introduces a thermal shock before it gets to assembly
- Not suited for fine pitch (due to uneven surface height) or high aspect ratio through-holes
- Standard SnPb HASL is not RoHS-compliant
- Pb-free HASL requires different assembly soldering parameters

OSP (Organic Solderability Preservative)

OSP is a thin organic coating, typically 5.9–11.8 μin (0.15–0.30 μm) thick, designed to prevent the oxidation of copper to maintain solderability over an extended period. Compared with other surface finish technologies, OSP is somewhat different. It uses a chemical process to produce an organic film on the bare copper surface which acts as a barrier to copper oxidation. OSP is organic, not metallic, and its cost is lower than most surface treatment technologies.

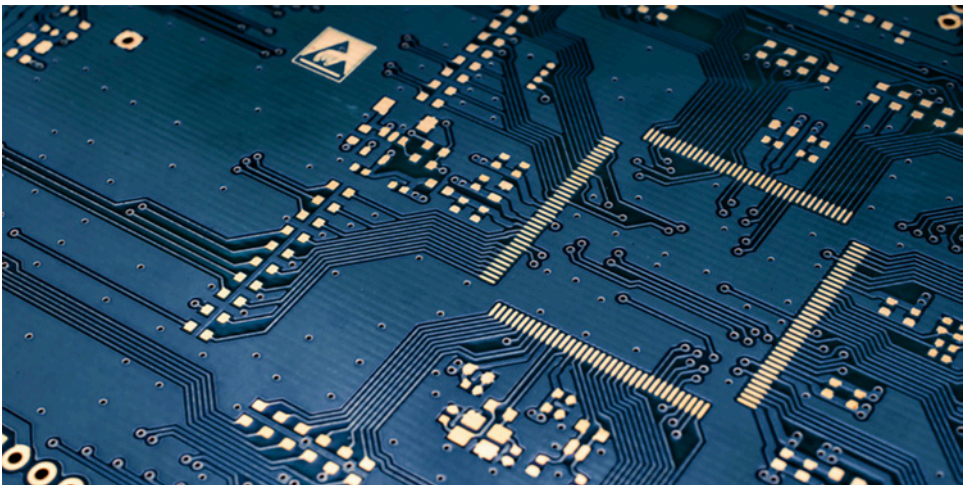


Figure 8.6: OSP PCB example.

As with all surface finishes, the primary purpose is to protect the solderable surfaces on the PCB from oxidation and to aid in assembly soldering. This process coats a very thin coating of an organic material that inhibits copper oxidation. It is so thin that it is nearly impossible to see and measure. The organic material is removed by the assembly flux. Boards that have been OSP coated will have bright copper pad coloration. OSP is specially designed for mixed metal applications, such as electroless nickel immersion gold (ENIG). The OSP selectively deposits on copper while leaving gold connectors or metallic heat sinks free of contamination.

These final finishes are widely used in the industry:

ENTEK CU-56: This finish is used for assemblies that will only go through a single reflow process. It is not being utilized much anymore because of the use of mixed-technology boards (SMT and through-hole).

ENTEK CU-106A: This is the most prevalent version primarily due to the ability to survive multiple thermal assembly operations. PCBs that have multiple surface finishes can use the CU-106A(X) finish.

Pros and Cons of OSP

Pros

- Inexpensive and typically used in fine pitch surface mount designs
- Environmentally friendly
- Provides a coplanar surface
- Lead-free (Pb-free)
- Mixed metal applications

Cons

- Not conducive to press-fit pin insertion
- Sensitive to handling
- Limited shelf life
- In-circuit test (ICT) difficulties
- Exposed copper after assembly
- Multiple reflow cycles and solder paste removal are problematic

Design Guidelines

Designing a printed circuit board surface finish is crucial for ensuring the reliability and performance of electronic devices. Here are some guidelines to consider.

Functionality and Application: Understand the requirements of your PCB in terms of functionality, environment, and application. Different surface finishes offer various advantages and are suitable for different scenarios. Considerations include:

- Environmental conditions
- Assembly process
- Electrical performance
- Component compatibility
- Reworkability
- Long-term reliability
- Cost and manufacturing constraints
- Corrosion resistance
- Solderability
- Compatibility with components
- Consideration of lead-free vs. lead (Pb) finishes
- Flatness and planarity
- Cost considerations
- Reliability
- RoHS compliance

By considering these guidelines and selecting an appropriate surface finish, you can ensure the reliability, performance, and manufacturability of your PCB design.

Table 8.1: Performance guidelines.

Finish	Handling Concerns	Hold Time Between Reflow Cycles	Shelf Life	Solderability
ENIG	Organics & fingerprints	No concerns	12 months	Excellent
ENIPIG / ENIPIG	Organics & fingerprints	No concerns	12 months	Excellent
Electrolytic Ni / Au	Organics & fingerprints	No concerns	12 months	Good
ImmAg	Sulfates & chlorides	No concerns	6 months	Excellent
ImSn	Organics & fingerprints, excessive heat cycles	24 hours	12 months	Excellent
HASL SnPb and Lead Free	No concerns	No concerns	12 months	Excellent
OSP	Organics & fingerprints	24 hours	6 months	Good

CHAPTER 8 REFERENCES

1. Per IPC-4556, nominal pad size of 0.060" x 0.060" (1.5 mm x 1.5 mm).
2. Per IPC-4553A, nominal pad size of 0.060" x 0.060" (1.5 mm x 1.5 mm).
3. Per IPC-4554, nominal pad size of 0.060" x 0.060" (1.5 mm x 1.5 mm).



Data Package Guidelines

Data Formats

There are many common data formats in use, depending on the design provider's software and preferences. PCB fabricators need to be able to work with all of the formats; however, the preferred formats at ASC are IPC-2581 and ODB++ because they provide a unified/consistent data format for all data—layers, drill, netlist, etc.—between the designer and the fabricator.

Common data formats:

- IPC-2581
- ODB++
- 247-D Gerber
- 274X Gerber
- 274X2 Gerber (not universally supported by all eCAD tools)
- 274X3 Gerber (not universally supported by all eCAD tools)
- IPC-D-356 Netlist
- Mentor Neutral Netlist
- HPGL
- DXF
- DWG
- NC Drill (Excellon file)
- PDF (for notes, specifications, and stackups)
- Excel (for stackups)

Part Drawing/Data Package Requirements

Drawing media includes paper, or any of the electronic data files listed above plus customer specifications, readme files, etc. Refer to IPC-D-325 and IPC-2610 for the requirements of different drawings.

The drawing/data package should include:

- Profile dimensions and tolerance for the board
- Hole-to-board edge dimension in two axes
- List of finished hole sizes and hole size tolerance
- Material type, thickness, and finished copper weight
- Applicable acceptability specifications as required
- Solder mask color, type, and number of sides
- Surface finish: SMOBC, white tin, electroless nickel/gold, etc.
- Nomenclature/legend color and number of sides

Array Drawings

Drawing media includes paper, Gerber, HPGL, PDF, Film Plot, AutoCAD.

If the board is to be supplied in array form, the array drawing shall have the following:

- Profile dimensions and tolerance for array
- Array edge to board edge (or hole) in two axes and orientation of pieces in the array
- Dimensional location, size, and tolerance of array tooling holes
- Dimensional location and size of array fiducials on copper and mask layer
- Breakaway slot detail and holding tab locations or scoring cross-sectional detail
- Number of "X-outs" allowed per array

Artwork

Common media include Gerber RS-274X, Gerber, and film; Gerber RS-274X is the preferred format versus Gerber RS-274D because it has embedded apertures and formatting within its file, which reduces translation/interpretation errors. Gerber RS-274D files require a separate aperture list. Film plots shall be used as received. As stated in the above Data Formats section, the preferred formats of ASC are IPC-2581 and ODB++.

Drill Data

Common media is Excellon format ASCII data. Paper coordinate lists and "bomb plotting" are to be avoided.

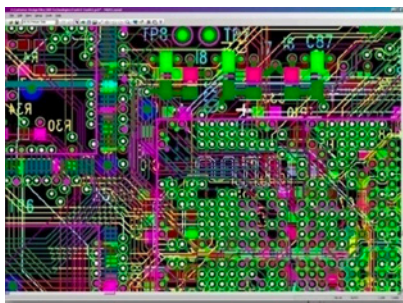


Figure 9.1: Part drawing/data examples.



Common Data Issues

Two types of data issues are frequently encountered. Minimization of these issues will decrease tool generation time. The following lists present the most common errors:

Critical: Can be substantial enough to stop the tooling process.

- Bad compression (cannot unzip)
- Missing FAB prints of files
- Formats not recognized
- FAB prints are not legible
- No README.TXT file
- Information does not match
- Outside manufacturing capabilities
- Missing Gerber files
- Missing drill files
- Missing aperture list
- Missing "D" codes
- Unclear aperture list
- Revisions do not match
- Netlist compare fails

Non-critical: Simply results in additional edits that need to be performed.

- Netlist not supplied
- No README.TXT
- Supplied design violates the OEM-provided specifications/documentation
- Supplied design violates the contract manufacturer provided specifications
- Violations on rules that supplier capability will improve yield/reduce cost



Cost Driver Summary

Low-cost Adders (<10% board cost)

Complex routing and scoring mean a small increase in process time, but a process is still required which will be driven by NC programming; it may limit tool life as a function of diameter. Thicker or thinner PCBs ($>0.093"$ [0.229 mm], $<0.030"$ [0.762 mm]) means a material cost variation but it is minimal. Via plug or button print requires a small process change to the screen-print mask dot.

Medium-cost Adders (10–25%)

Regarding drilled hole quantity, there is a cost adder for high-density design-driven hole count and process time. With smaller drilled hole size, the small drill diameter ($<0.010"$ [0.254 mm]) limits throughput and stack height. Embedded resistors with Ohmega/Ticer technology will need additional core testing and finished board verification. Non-FR-4 materials, like PTFEs, can be 10 to 20 times the FR-4 cost, and material cost is generally 25–50% of the board cost. Edge plating will have additional processes required prior to plating.

High-cost Adders (>25% board cost)

Advanced technologies may become “science projects” with industry non-standard processing or materials, or “bleeding edge” technology ($<0.003"$ [0.076 mm] L/S, 1:1 aspect ratio microvias, 0.4 mm BGA technology, etc.). Sequential lamination and complex via structures, as well as metal core or external heat sink requirements, also add cost.

If layer count will be >30 layers, the yield impact can be significant. Combination/hybrid material sets can be high-cost adders. Regarding material/panel utilization, test coupons and board size can greatly reduce panel utilization; array configurations reduce panel utilization with unusable real estate; and there may be limited availability of panel sizes.



Selective plating means multiple surface finishes or multiple thicknesses, complex processing requirements (e.g., masking), and the yield risk of combining non-standard processes. There are also cost adders for line width and spacing below industry standards for any given copper weight:

- ½-ounce copper: 0.003"/0.003" (0.076 mm x 0.076 mm)
- 1-ounce copper: 0.004"/0.004" (0.102 mm x 0.102 mm)
- Ultra-high-density interconnects
- Sub 1 mil line/space (25 and 15 micron)

Understanding the cost drivers in PCB fabrication and early engagement between designer and fabricator are crucial elements that lead to cost-effective design success.

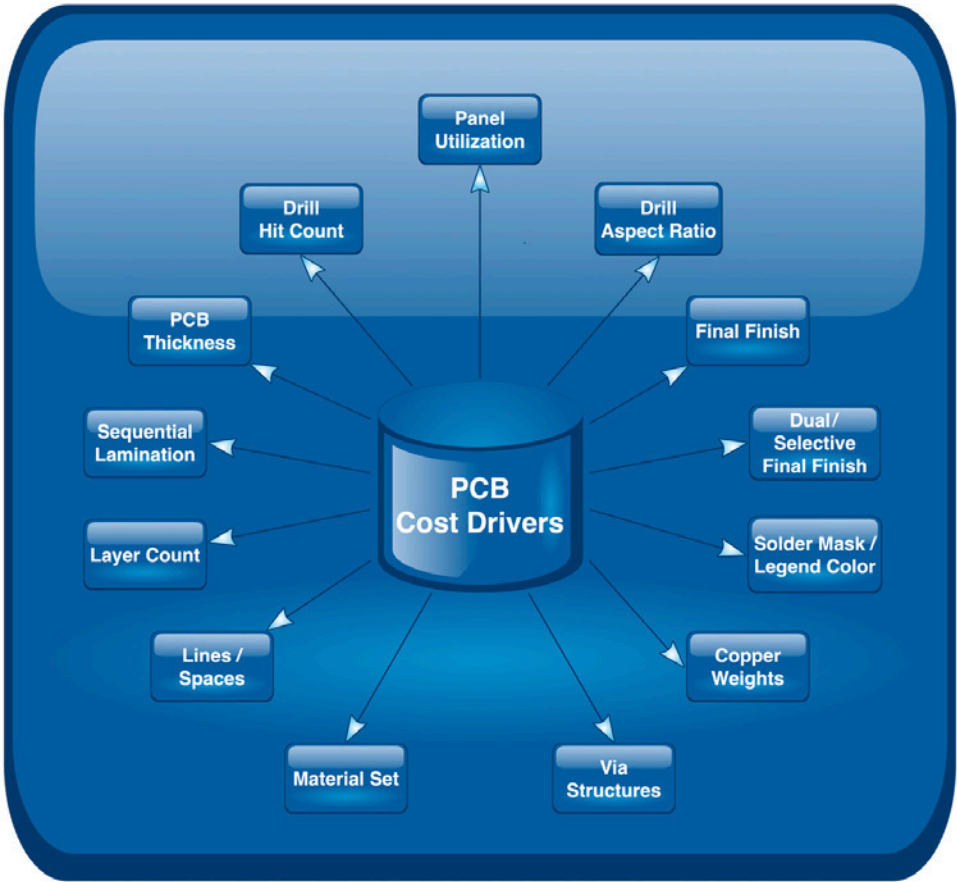


Figure 10.1: PCB cost drivers.

Conclusion

The need for Design for Manufacturability (DFM) in the product development process is paramount for achieving efficient and cost-effective manufacturing, particularly with complex products like printed circuit boards. By integrating DFM principles into the design phase and collaborating early with your PCB fabricator, designers can address potential challenges early on, streamline the production process, and enhance the overall manufacturability of the product. Here are key points highlighting the importance of DFM:

Cost Efficiency: DFM focuses on optimizing designs for ease of manufacturing. Addressing manufacturability issues up front helps minimize production costs by reducing the likelihood of rework, scrap, or the need for complex manufacturing processes.

Time Savings: Design modifications made during the early stages of product development are more time-efficient than making changes later in the process. DFM can accelerate time-to-market by avoiding delays caused by manufacturing hurdles discovered late in the design cycle.

Quality Improvement: Considering manufacturing constraints from the outset contributes to improved product quality. Designs that are manufacturable with fewer defects lead to more reliable and durable products, ultimately enhancing customer satisfaction.

Enhanced Collaboration: DFM encourages collaboration between design and manufacturing teams. Early engagement between these teams facilitates the exchange of valuable insights, ensuring that designs are not only innovative but also practical to produce.

Supply Chain Optimization: DFM aids in optimizing the supply chain by selecting materials and components that are readily available, cost-effective, and meet the necessary quality standards. This contributes to a more resilient and efficient manufacturing ecosystem.

Scalability: Designs that are inherently manufacturable are more scalable. Whether producing a small batch or scaling up for mass production, a manufacturable design is adaptable to different production volumes.

Reduced Environmental Impact: Efficient manufacturing processes, enabled by DFM, often result in reduced waste and resource consumption. This aligns with sustainable and environmentally friendly manufacturing practices.

Compliance and Standards: DFM helps ensure that products meet regulatory and industry standards. Designing with manufacturing considerations in mind aids in compliance with safety, quality, and performance requirements.

In summary, the integration of Design for Manufacturability is a strategic approach that not only benefits the manufacturing process but also contributes to the overall success of a product in terms of cost, time, quality, and sustainability. It is an essential practice for companies aiming to create innovative, competitive, and efficiently manufacturable products in today's dynamic market.



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About American Standard Circuits

In the world of DFM, American Standard Circuits and ASC Sunstone Circuits come together as a unified PCB manufacturing powerhouse with over 50 years of combined experience. Together, they offer a comprehensive range of solutions and excel in producing ultra HDI, rigid, metal-backed, RF/microwave, flex, and rigid-flex PCBs for various industries, ranging from industrial to aerospace/defense.

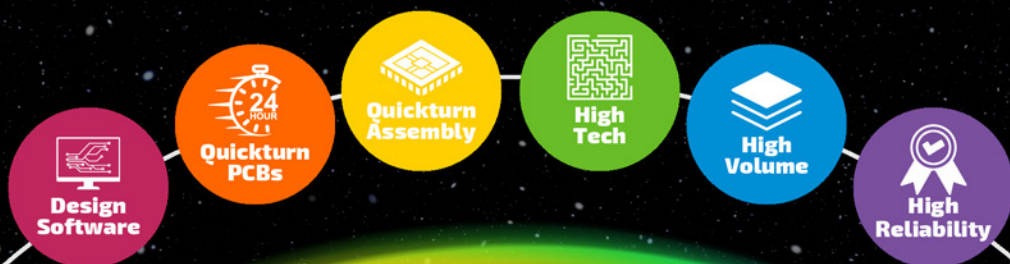
This strategic alliance represents their dedication to delivering exceptional products and simplifying the PCB procurement process. By merging their strengths, they have set a groundbreaking standard in PCB manufacturing, providing engineers with the ultimate PCB solution for everything from design and prototypes to large-scale production, all based on the principles of Design for Manufacturability.

The company's impressive certifications and advanced technology expertise reflect its commitment to excellence. They prioritize unparalleled customer support, ensuring that clients' needs are always met with precision and care.



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